

Programme «Pedagogy»

**PEDAGOGICAL POTENTIAL OF LARGE
LANGUAGE MODELS IN IMPLEMENTING
PERSONALIZED FEEDBACK SYSTEMS IN
MATHEMATICS LESSONS**

(Master's Thesis)

Student of the Programme
«Pedagogical Practice in a Changing Environment»,
TFK-124, Aidin Biibosunov

Supervisors: Prof. Dr. U. Mambetakunov,
I. Ninenko

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1 Introduction

1.1 Relevance

Feedback is widely recognized as one of the most powerful influences on student learning and achievement. A landmark meta-analysis Wisniewski et al. (2020) synthesized findings across 435 studies and reported an overall effect size of $d = 0.48$ for educational feedback, rising to $d = 0.99$ for high-information feedback that provides specific guidance on learning processes. Earlier synthesis work Arnold (2011) across more than 800 meta-analyses placed feedback among the top instructional strategies with an effect size of $d = 0.73$, underscoring its centrality to effective teaching and learning. These figures situate feedback not as a supplementary pedagogical tool but as a foundational mechanism through which students develop understanding, correct misconceptions and advance toward mastery.

Despite this compelling evidence, mathematics classrooms in practice frequently fall short of realizing the potential of effective feedback. In an extensive review of formative assessment research, Black and Wiliam (1998) documented a persistent gap between what research recommends and what teachers are able to deliver: feedback in mathematics lessons is infrequent, overly general, and overwhelmingly limited to right-or-wrong evaluations that tell students what they got wrong without explaining why or how to improve. Hattie and Timperley (2007) elaborate the mechanism behind this effect. They distinguish four feedback levels: task, process, self-regulation, and self. Their analysis shows that process- and self-regulation-level feedback produce the largest learning gains. The reasons for this gap are clear: too many students per class, not enough time, packed curricula, and teachers who aren't fully trained in giving feedback all make it nearly impossible to give every student useful, personal feedback on a regular basis. This is especially acute in secondary school mathematics, where conceptual difficulty accelerates and misconceptions left unaddressed can compound across topics and years.

The emergence of Large Language Models (LLMs) offers an unprecedented technological opportunity to bridge this feedback gap at scale. Recent empirical studies demonstrate that LLM-based systems can perform complex pedagogical tasks that were previously limited to trained human assessors. Liu et al. (2024)

showed that AI-assisted automated grading of handwritten university-level mathematics exams achieves human-level accuracy across a range of problem types. This capability builds on earlier automated scoring research. Ramesh and Sanampudi (2022) review the development of neural automated essay scoring systems and establish the technical lineage from which LLM-based grading descends. Subsequent work has confirmed and extended this finding: Chu et al. (2024) demonstrated that a prompt-optimized LLM grading framework achieves human-level accuracy across open-ended secondary school responses. Chen and Wan (2025) demonstrated that LLMs can assign partial credit to students' explanations with reliability comparable to expert human graders. Kortemeyer (2023) adds direct evidence from mathematics specifically. His study shows that generative AI can assist students in solving mathematics problems at a level approaching that of expert tutors. Pardos and Bhandari (2024) found that ChatGPT-generated help produces learning gains equivalent to human tutor-authored help on mathematics skills. A recent PRISMA-based review of 45 studies confirms that AI-driven personalization consistently improves learning outcomes when embedded within coherent instructional frameworks (Wafik et al., 2025). This is consistent with the broader systematic review of AI-based assessment by Zhai et al. (2021). They identify automated feedback as the most pedagogically impactful application of AI in educational settings. These findings collectively suggest that LLM-based assessment systems are not merely technically feasible but pedagogically capable of delivering the kind of specific, process-oriented feedback that research identifies as most effective.

However, technological capability alone is insufficient to guarantee educational benefit. Kasneci et al. (2023) emphasize that the integration of LLMs into educational settings must be grounded in sound pedagogical theory to avoid superficial or even counterproductive implementations. Schaaf et al. (2026) demonstrate that the mere frequency of use of an intelligent tutoring system does not, by itself, predict learning gains in secondary mathematics. What matters is how the system is embedded within the broader pedagogical structure of instruction. These findings caution against naive enthusiasm and reinforce the need for principled, theory-driven design. Internationally, OECD (2026) and UNESCO (2023) have both called for human-centered approaches to artificial intelligence in education that are grounded in established pedagogical principles and that maintain teacher professional authority.

This thesis addresses precisely the intersection between emerging technological capability and established pedagogical theory by investigating the pedagogical potential of LLM-based assessment systems in secondary school mathematics education. Rather than treating LLMs as a general-purpose educational technology, the research conceptualizes them as a targeted instrument for addressing a specific pedagogical problem: the chronic inability of mathematics teachers to provide timely, process-oriented, and individualized feedback to every student in every lesson. The AI Grader system (AIsuluu) developed and evaluated in this study is therefore designed not to automate teaching itself, but to automate the most time-intensive component of formative assessment practice. Grounded in theories of formative assessment, personalized learning, differentiated instruction, and Lev Vygotsky's Zone of Proximal Development, the system aims to transform a technically capable tool into a genuinely pedagogically effective intervention. By redistributing routine assessment labor to the machine, the approach seeks to free teachers to concentrate on the relational, motivational, and instructional dimensions of teaching that cannot be replicated through automation. The study further investigates the conditions under which such a redistribution of labor between human and machine can be considered pedagogically sound, practically feasible, and institutionally appropriate within the specific context of secondary school mathematics education in Kyrgyzstan.

1.2 Goal

To investigate the pedagogical potential of Large Language Models in implementing personalized feedback systems in mathematics lessons at the secondary school level.

1.3 Objectives

1. To analyze existing literature on formative assessment, personalized learning, differentiated instruction, and the Zone of Proximal Development as they apply to AI-assisted mathematics education.
2. To develop a pedagogically grounded conceptual framework for an LLM-based assessment system ("AI Grader - AIsuluu") that integrates established educational theories.

3. To design three concrete implementation scenarios for deploying the AI Grader in secondary school mathematics classrooms.
4. To validate the proposed scenarios through a focus-group with experienced mathematics teachers.

1.4 Object of Study

The process of feedback provision in secondary school mathematics education.

1.5 Subject of Study

The use of Large Language Models to implement personalized feedback systems in secondary mathematics assessment.

1.6 Hypothesis

LLM-based automated grading systems, when grounded in formative assessment theory and Vygotsky's Zone of Proximal Development, can effectively reduce the feedback gap in secondary school mathematics by providing timely, personalized, and process-oriented feedback at scale while maintaining teacher authority through a human-in-the-loop model.

1.7 Research Methods

The research employs three primary methodological approaches:

1. Systematic literature review of empirical and theoretical sources on formative assessment, personalized learning, AI in education, and LLM-based grading.
2. System design based on pedagogical framework synthesis, resulting in the conceptual architecture of the AI Grader and three implementation scenarios.
3. Qualitative research using a focus-group with experienced secondary school mathematics teachers to validate and refine the proposed scenarios.

1.8 Structure of the Thesis

Chapter 1 introduces the research problem, establishes the relevance of the study, defines the research objectives and hypothesis, and outlines the methodological approach.

Chapter 2 presents the theoretical framework of the study. It reviews literature on formative assessment, personalized learning, differentiated instruction, scaffolding pedagogy, self-regulated learning, the Zone of Proximal Development, and human-in-the-loop models in AI-supported education.

Chapter 3 describes the research methodology and the conceptual design of the AI Grader system, including the three implementation scenarios and the focus group procedure.

Chapter 4 presents the empirical findings from the focus group, including content analysis and thematic analysis of teacher responses.

Chapter 5 discusses the findings in relation to the theoretical framework and identifies implications for AI-assisted mathematics assessment.

2 Theoretical Part: Pedagogy in Mathematics Education

This chapter presents a systematic analytical review of the scientific literature underpinning the pedagogical framework for LLM-based assessment in secondary school mathematics. The review is organized thematically rather than chronologically, moving from foundational challenges in mathematics feedback (Section~2.1) through the theoretical frameworks that inform effective pedagogy (Section~2.2–Section~2.8) to a synthetic integration that constitutes the conceptual architecture of the proposed AI Grader system (Section~2.9).

2.1 The Feedback Gap in Mathematics Education

Feedback is among the most thoroughly evidenced influences on academic achievement. In a meta-analysis of 435 studies, Wisniewski et al. (2020) reported an overall effect size of $d = 0.48$ for educational feedback, with high-information, process-oriented feedback reaching $d = 0.99$, placing it in the highest tier of instructional interventions. In a broader synthesis of over 800 meta-analyses, Arnold (2011) confirmed feedback's position among the top ten most powerful influences on student learning, with a synthesized effect size of $d = 0.73$. Crucially, both analyses emphasize that the magnitude of feedback effects depends heavily on its quality: feedback that tells students whether an answer is correct produces modest gains, while feedback that explains why an error occurred and guides the student toward correction produces dramatically larger ones. Mathematics education involves both procedural and conceptual knowledge, and the two are tightly intertwined. A misunderstanding at one level can easily spread into systematic errors at later levels. For this reason, the difference between low and high-quality feedback is not merely academic. It has real educational consequences.

Yet in practice, mathematics classrooms persistently fall short of delivering the quality of feedback that research recommends. In a landmark review of over 250 studies on formative assessment, Black and Wiliam (1998) documented a pervasive pattern: feedback in school mathematics is infrequent, insufficiently specific, and disproportionately focused on correctness rather than process. Black and Wiliam (2009) later theorized the mechanisms behind this gap, arguing that sustained improvement requires systemic changes to classroom assessment culture rather than individual teacher effort alone. Students routinely receive grades or ticks

without any accompanying explanation of their reasoning errors or guidance on how to revise their thinking. The structural drivers of this gap are well understood: a secondary school mathematics teacher managing classes of thirty to forty students across multiple sections must process hundreds of homework submissions and assessments each week. Providing even a single substantive written comment per student per assignment would require hours of additional work that the instructional day cannot accommodate. As a result, feedback that research identifies as most effective, namely specific, timely, actionable, and process-focused, becomes practically unsustainable at scale.

The consequences of chronically poor feedback quality are measurable and cumulative. When students make errors in mathematical problem-solving and receive only binary right/wrong evaluations, they are denied the information needed to identify the source of their misconception, to understand what a correct approach looks like, and to regulate their own learning accordingly. Over time, unaddressed misconceptions compound: a student who does not receive feedback on a sign error when solving linear equations will carry that error into quadratic equations, systems of equations, and beyond. Research by Black and Wiliam (1998) documents that this is particularly detrimental for slower-processing learners, who depend more heavily on instructional feedback because they are less likely to infer correct procedures from observing peers. Lema and Maro (2018) document structurally identical feedback deficits in Tanzanian secondary mathematics classrooms, confirming that the gap is systemic rather than context-specific. The feedback gap thus contributes directly to the widening of achievement gaps within and across mathematics classrooms. Shute (2008) developed a comprehensive taxonomy of feedback. It classifies feedback by specificity, timing, and cognitive level. This taxonomy provides the analytical framework used to design the AI Grader's process-level feedback prompts.

Large Language Models represent a qualitatively new technological capacity to address this gap. Liu et al. (2024) demonstrated that an LLM-based system for grading handwritten university mathematics exams achieved accuracy comparable to human graders across a range of problem types, including multi-step proofs and applied calculations. Chen and Wan (2025) showed that LLMs can assign partial credit to students' written explanations (in domain-specific tasks such as physics) at levels of inter-rater reliability equivalent to trained human scorers. Biibosunov

(2025) introduced the AI Grader system (AIsuloo) which integrates OCR-based digitization of handwritten submissions with LLM-generated personalized feedback at scale. These developments collectively indicate that the technical barrier to high-frequency, high-quality automated feedback in mathematics has been substantially reduced. The remaining challenge (which this thesis addresses) is translating this technical capability into pedagogically effective practice grounded in established educational theory.

2.2 Personalized Learning Theory

Personalized learning represents a fundamental shift in educational philosophy: from the industrial-era model of standardized instruction delivered uniformly to cohorts of students, toward an approach that recognizes and actively responds to individual learner variability. Patrick et al. (2013) define personalized learning as "tailoring learning for each student's strengths, needs, and interests — including enabling student voice and choice in what, how, when, and where they learn — to provide flexibility and supports to ensure mastery of the highest standards possible". This definition emphasizes two interrelated dimensions: the diagnostic dimension (understanding where each learner currently is) and the responsive dimension (adapting instruction and support to close the gap between current and target performance). In mathematics education, this means accounting for differences in students' prior knowledge, how quickly they process information, how familiar they are with mathematical representations, and how motivated they are to learn.

The theoretical roots of personalized learning extend back to the sociocultural theory of Vygotsky (1978) and Bloom's mastery learning model, both of which share the conviction that virtually all students can achieve high standards given appropriate time, support, and feedback. Bloom (1984), known as the "2-Sigma Problem" paper, is the direct empirical source for Bloom's mastery learning claim. Contemporary formulations of personalized learning add technological mediation as a central mechanism: digital tools can collect granular data about each student's performance, identify patterns of strength and difficulty, and use these patterns to adapt the learning environment in real time. In mathematics education, this means moving beyond aggregate class-level data toward individual profiles that capture not just scores but the specific types of errors a student makes, the strategies they employ, and the kinds of support that have proven most effective for them, as

Patrick et al. (2013) emphasize.

The relationship between personalized learning and feedback is particularly direct. High-quality personalized feedback is simultaneously a product of a personalized learning system. It responds to what this specific student did in this specific problem, and serves as a mechanism that drives further personalization by generating richer data about individual learning. Wisniewski et al. (2020) note that the highest-effect feedback is precisely the feedback that is most personalized: it addresses the individual's specific errors, misconceptions, and knowledge gaps rather than providing generic guidance. This creates a virtuous cycle in which personalized feedback both reflects and deepens the personalization of the learning experience. LLM-based assessment systems are uniquely positioned to operationalize this cycle at scale, generating individualized feedback for each student on each submission in ways that manual grading cannot achieve across large classes.

Empirical evidence for the effectiveness of personalized learning in mathematics is growing, though the research landscape is nuanced. Henkel et al. (2024) conducted a rigorous study of an AI tutor deployed in secondary mathematics classrooms in Ghana and found significant positive effects on student achievement, with effect sizes comparable to high-dosage human tutoring. Earlier work Pelkola et al. (2018) demonstrated that Bloom's mastery learning can be operationalized through automated online assessment in university mathematics, constituting a direct technological precursor to the AI Grader concept. Pardos and Bhandari (2024) reported equivalent learning gains from ChatGPT-generated mathematics help and human tutor-authored help, suggesting that LLM personalization can reach parity with human tutoring on targeted skills. A systematic review of 78 studies from 2019 to 2024 confirms that AI-mediated personalized learning is increasingly validated across educational contexts, with effectiveness dependent on quality of pedagogical integration, as Bayly-Castaneda et al. (2024) report. These findings are instructive: they demonstrate that AI-mediated personalized learning is not merely theoretically promising but empirically validated as effective when implemented thoughtfully. The key condition, consistently emphasized by Kasneci et al. (2023), is that personalization must be embedded within a coherent pedagogical design rather than delivered as a stand-alone technological feature. A large-scale study Pane et al. (2015) across 62 schools found that students in personalized learning environments gained significantly more in mathematics than matched peers in

traditional instruction. This provides rigorous causal evidence for the effectiveness of the approach.

2.3 Differentiated Instruction

Differentiated instruction is the practical classroom manifestation of personalized learning philosophy. Tomlinson (2017) defines it as a framework in which teachers proactively plan varied approaches to what students need to learn, how they will learn it, and how they will demonstrate their learning, in order to increase the likelihood that each student will learn as much as possible. The framework identifies four dimensions along which instruction can be differentiated: content (what students learn), process (how students engage with and make sense of new material), product (how students demonstrate understanding), and learning environment (the conditions under which learning takes place). In secondary school mathematics, differentiation across these dimensions might involve assigning problems of varying complexity, providing worked examples for some students while offering only prompts for others, allowing different representational formats for demonstrating understanding, and organizing flexible small-group instruction around shared learning needs.

The pedagogical rationale for differentiated instruction is grounded in the recognition that student readiness, interest, and learning profile vary substantially within any given classroom. Variation that a uniform instructional approach inevitably fails to address. In mathematics specifically, prior knowledge is a particularly powerful moderator of learning: a student who has not yet consolidated fraction operations cannot productively engage with algebraic manipulation, regardless of how well the algebraic content is taught. Differentiated instruction addresses this by ensuring that instructional tasks and supports are calibrated to the actual knowledge state of each learner, not to an assumed class average, as Tomlinson (2017) argues. Hamid and Ahmad (2025) argue that AI-enhanced pedagogical systems in secondary mathematics education must incorporate differentiated instruction principles to remain effective across the full range of student ability levels, and propose a conceptual framework that integrates AI capabilities with core differentiation strategies. A systematic review Smale-Jacobse et al. (2019) provides the empirical basis for this claim, finding effect sizes of $d = 0.50 - 0.74$ for differentiated instruction in secondary education when responsive to ongoing

assessment data. Insorio and Librada (2024) confirm these effects in a secondary mathematics setting directly comparable to the Kyrgyz context studied here.

The practical challenge of differentiated instruction is the same challenge that afflicts personalized learning more broadly: it is immensely demanding of teacher time and cognitive bandwidth. Designing tiered tasks for three or four readiness levels takes considerable effort. Managing the logistics of simultaneous different activities in a single classroom adds further complexity. On top of this, teachers must provide differentiated feedback to students working on different problems. All of this requires sustained professional effort that most teachers, operating under real-world constraints of time and class size, can sustain only irregularly, as Black and Wiliam (1998) document. This is precisely the point at which AI-assisted assessment offers meaningful pedagogical leverage. One of the biggest challenges in differentiation is the time it demands from teachers. Tasks such as diagnosing current readiness, generating appropriately tiered feedback, and identifying students who need additional scaffolding are all time-intensive. LLM-based systems can automate these components. In doing so, they can make differentiated instruction a routine classroom practice rather than an occasional exceptional effort.

2.4 Formative Assessment and Feedback

Formative assessment is not defined by particular instruments or timing but by the pedagogical function it serves: the use of assessment evidence to adjust instruction and learning in ways that better meet student needs. In a foundational review, Black and Wiliam (1998) established that improving formative assessment practices produces among the largest learning gains ever documented for educational interventions, with effect sizes ranging from 0.4 to 0.7 standard deviations. This is equivalent to moving an average student from the 50th to the 70th percentile. The review identified five key formative assessment strategies: sharing learning intentions and success criteria, engineering effective classroom discussions and tasks, providing feedback that moves learners forward, activating students as learning resources for each other, and activating students as owners of their own learning. Not all types of feedback are equally effective. The most powerful kind is feedback that moves learners forward. This means providing specific, timely, and actionable information about the gap between a student's current and desired performance.

At the same time, this type of feedback is the most challenging to implement consistently. Wiliam (2011) extends this framework into a practical model for continuous classroom implementation. He argues that formative assessment must be embedded in daily instructional routines. Reserving it for discrete testing events reduces its effectiveness considerably.

Leahy et al. (2005) extended this framework with the concept of "minute-by-minute" and "day-by-day" formative assessment, in which assessment is not a periodic event but a continuous, embedded feature of instruction. In this model, every interaction between student and teacher, every piece of work submitted, and every mathematical task completed is potentially formative. It generates evidence that the teacher can use to adjust instruction, that the student can use to self-regulate learning, and that both can use to identify the next productive step. The challenge is operational: creating the conditions for continuous formative assessment requires feedback infrastructure that can process student work quickly enough to influence the next instructional moment, not days or weeks later when the pedagogical window has closed.

LLM-based grading systems address this operational challenge directly. Bii-bosunov (2025) describes an AI Grader implementation in which students receive personalized feedback on homework submissions within approximately one hour of submission, a cadence that approaches the "day-by-day" frequency that Leahy et al. (2005) identify as optimal for formative assessment. This rapid feedback cycle transforms the temporal structure of formative assessment: instead of a teacher reviewing forty homework papers over an evening and returning them with comments several days later, students receive specific feedback while the mathematical content is still active in their working memory. Simultaneously, the teacher receives a class-wide analysis of misconceptions and error patterns that can directly inform the design of the following day's lesson, realizing in operational terms what Black and Wiliam (1998) describe as the core mechanism of effective formative assessment. Causal evidence for this mechanism comes from Wang and Sperling (2020), who found that feedback quality directly predicts mathematics performance at secondary level independently of prior achievement.

The quality dimension of formative feedback is equally important. Wisniewski et al. (2020) distinguish between feedback that focuses on the task level (whether the answer is correct), the process level (what reasoning strategy was used

and where it broke down), the self-regulation level (how the student can monitor and adjust their own approach), and the self level (global evaluations of the student as a person). Their meta-analysis found that process- and self-regulation-level feedback produce significantly larger learning gains than task- and self-level feedback. Nicol and Macfarlane-Dick (2006) operationalize this directly. They propose seven principles of good feedback practice that simultaneously improve performance and develop self-regulatory capacity. This dual function makes formative feedback the central engine of self-directed learning. Well-designed LLM feedback prompts, grounded in this taxonomy, can specifically target process-level explanation. This means identifying not just that a student got the wrong answer, but which step in the solution procedure went wrong and why. In this way, each automated feedback instance can maximize its formative impact.

2.5 Zone of Proximal Development

Lev Vygotsky's concept of the Zone of Proximal Development (ZPD) is one of the most influential constructs in twentieth-century educational psychology and remains a cornerstone of contemporary instructional design. Vygotsky (1978) defined the ZPD as "the distance between the actual developmental level as determined by independent problem solving and the level of potential development as determined through problem-solving under adult guidance or in collaboration with more capable peers". The ZPD represents the productive space of learning: tasks that fall within the ZPD are those that a learner cannot yet accomplish independently but can accomplish with appropriate support. Tasks below the ZPD involve already-mastered material and produce no new learning; tasks above the ZPD exceed what support can bridge and produce frustration and disengagement rather than growth.

The pedagogical implications of the ZPD for mathematics education are profound and specific. At the level of task design, the ZPD demands that teachers identify each student's current level of independent competence in the mathematical domain being taught (not an easy task in a class of forty heterogeneous learners) and assign tasks that challenge the student without overwhelming them. This requires ongoing formative assessment data, which is precisely what AI-assisted grading systems are designed to generate. At the level of feedback design, the ZPD implies that feedback should bridge the gap between what a student can currently do and

what they are being asked to do: it should not simply tell students the correct answer (which would bypass the zone entirely) but should provide the specific conceptual scaffold that enables the student to work across the gap to understanding.

In the context of LLM-based assessment, the ZPD provides a principled criterion for evaluating feedback quality. Feedback that merely marks errors as incorrect without explanation operates below the level of the ZPD. It neither diagnoses the zone nor provides scaffolding across it. Effective feedback does more than mark an answer as wrong. It identifies exactly where in the student's reasoning the error occurred, names the misconception behind it, and offers a guiding question or hint to steer the student toward the correct approach. This is precisely the kind of feedback that operates within the ZPD, as Vygotsky envisioned. Margolis (2020) operationalizes this principle by distinguishing ZPD-aligned instruction from Bruner's scaffolding, providing a model for structuring tasks within each learner's zone of productive challenge. Shabani et al. (2010) extend this to dynamic assessment: the ongoing evaluation of what a learner can achieve with support, which maps directly onto the AI Grader's continuous readiness profiling. This distinction translates directly into LLM prompt design: effective AI Grader prompts must specify not just that the system should identify errors but that it should explain the reasoning behind them and provide scaffolded guidance toward correct thinking.

2.6 Scaffolding Pedagogy

Wood et al. (1976) introduced the concept of instructional scaffolding in their foundational study of adult-child tutoring interactions. Drawing on Vygotsky's theoretical framework, they used the architectural metaphor of scaffolding to describe the temporary, adjustable support structure that a more knowledgeable other provides to enable a learner to accomplish tasks beyond their current independent capability. The key features they identified are contingency (support is responsive to the learner's current performance), fading (support is gradually withdrawn as competence grows), and transfer of responsibility (the learner progressively takes over control of the task). They constitute what has become the standard model of effective scaffolded instruction. van de Pol et al. (2010) confirm in a systematic review that contingency is the most critical of these features. Support that does not adapt to demonstrated learner progress fails to qualify as genuine scaffolding. Such static support consistently produces weaker learning outcomes.

In mathematics education, scaffolding takes many forms: worked examples that model the problem-solving process; guiding questions that direct attention to a critical step without providing the answer; visual representations such as number lines, area models, or graphs that make abstract relations concrete; and incremental problem sequences in which each problem provides partial structure for the next. Scaffolding reduces the cognitive load associated with the most difficult aspects of a task. This frees up cognitive resources for the aspects that are currently at the frontier of the learner's competence. At the same time, scaffolding preserves enough challenge to ensure that learning still occurs, consistent with the accounts of Wood et al. (1976) and Tomlinson (2017). Effective scaffolding in mathematics does not simplify the mathematics; it adjusts the support structure around the mathematics.

The translation of scaffolding principles into LLM-based feedback design is both natural and demanding. Natural, because LLMs can generate contextualized, problem-specific scaffolds. For example, they can provide a hint for a specific problem, or a worked example of a closely related problem. They can also offer a guiding question that targets the exact step where a student's reasoning went wrong. Producing this kind of individualized support manually for every student would simply be impractical. Demanding, because the fading and contingency properties of effective scaffolding require that the system recognize when a student has successfully worked across the gap and adjust the level of support accordingly, rather than providing the same level of scaffolding regardless of demonstrated competence. This dynamic scaffolding capability is an area of active development in LLM-based educational systems Kasneci et al. (2023) and Liu et al. (2024) and represents one of the key design challenges for the AI Grader framework proposed in this thesis. Pea (2004) anticipates this possibility in his analysis of distributed scaffolding. He argues that computational tools can extend the scaffolding function beyond the one-to-one tutor relationship. This makes scaffolded support available in whole-class and asynchronous learning environments.

2.7 Self-Regulated Learning

Self-regulated learning (SRL) theory, developed extensively by Zimmerman (2002) and others, describes the processes through which learners take ownership of their own learning by setting goals, monitoring their progress toward those

goals, evaluating their performance against criteria, and adjusting their strategies in response to feedback. They articulate a three-phase cyclical model: the forethought phase (goal setting, planning, motivation), the performance phase (strategy use, self-monitoring, self-control), and the self-reflection phase (self-evaluation, causal attribution, adaptations for future performance). Pintrich (2000) complements this account by demonstrating that goal orientation is a key mediator of self-regulatory behavior. Students who pursue mastery goals rather than performance goals engage more deeply with feedback. This has direct implications for how AI-generated feedback should be framed to support intrinsic motivation. In mathematics education, self-regulation is particularly important because mathematical problem-solving requires metacognitive awareness. The ability to recognize when a strategy is not working, to step back from a dead end, and to apply alternative approaches.

The relationship between feedback and self-regulation is reciprocal and reinforcing. As Zimmerman (2002) argue, effective formative feedback supports the development of self-regulation by providing students with information they can use to calibrate their own understanding, to identify the gap between their current performance and the target, and to select more effective strategies. Wisniewski et al. (2020) identify process- and self-regulation-level feedback as among the highest-impact feedback types. This type of feedback explicitly guides students on how to monitor and adjust their own learning processes. Conversely, students who are more self-regulated derive greater benefit from feedback because they are better equipped to use it: they are more likely to read feedback carefully, to understand its implications for their approach, and to modify their behavior accordingly.

AI-assisted assessment systems can play a significant role in supporting self-regulation development, provided they are designed with this goal in mind. Feedback that not only identifies errors but explicitly prompts students to reflect on their problem-solving process—for example, "What was your reasoning at this step? What would you do differently?"—activates the self-reflection phase of Zimmerman's cycle and develops metacognitive habits over time, in line with the model of Zimmerman (2002). Darr and Fisher (2005) demonstrate how self-regulation habits can be cultivated in secondary mathematics through reflection tasks embedded in regular lessons. The same logic underpins the redo mechanism in Scenario 1 and the self-reflection prompts in Scenario 2 described in Chapter 3. Systems that provide students with visible records of their own feedback history allow them to

track patterns in their errors, observe their own progress, and identify persistent areas of difficulty. These features support the self-monitoring function that characterizes self-regulated learners. Such design principles are directly applicable to the AI Grader system and inform the structure of the feedback templates.

2.8 Human-in-the-Loop Model and Adaptive Learning Technology

The human-in-the-loop (HITL) model represents a design philosophy for AI systems in which human judgment, oversight, and decision-making remain central to the system's operation rather than being replaced by automation. In educational contexts, the HITL model is not merely a technical architectural choice but a pedagogical and ethical imperative. Kasneci et al. (2023) identify teacher authority and professional judgment as non-negotiable components of responsible AI integration in education: AI systems should augment teacher capacity rather than substitute for teacher expertise, and should operate under conditions of transparency and teacher control. Luckin et al. (2016) make a complementary argument in their foundational policy report. They contend that AI in education should function as an intelligence amplifier for teachers rather than as an autonomous agent. This framing aligns directly with the human-in-the-loop design philosophy adopted in this thesis. Holmes and Tuomi (2022) extend this to an ethical framework. They identify transparency, accountability, and avoidance of harm as the three conditions under which AI assessment systems can be considered pedagogically responsible. These are precisely the conditions that the HITL model is designed to satisfy. OECD (2026) and UNESCO (2023) reinforce this position in their institutional guidance documents, calling for AI deployments in education that maintain clear human accountability for assessment decisions and that position AI as a tool that augments rather than replaces teacher judgment.

The practical implementation of the HITL model in an AI Grader system involves designing explicit checkpoints at which teacher review and intervention are not just permitted but structurally required. In the context of the three implementation scenarios proposed in this thesis, AI-generated feedback is always visible to the teacher before or alongside its delivery to students. The teacher also has the ability to edit, add to, or override any AI-generated comment. Additionally, AI-generated class-wide analysis of errors is presented as data for teacher interpretation, rather than as an autonomous instructional decision. This architecture

preserves the teacher's role as the primary educational authority while dramatically reducing the time burden associated with individual feedback provision. This combination makes the HITL model both pedagogically sound and practically viable. Yang (2026) gives this concrete form by proposing four teacher roles under HITL: monitor, evaluator, corrector, and curriculum architect. Each mapping directly onto teacher actions described in the three AI Grader scenarios. Empirical support comes from Tian et al. (2025), whose co-design study with 19 K-12 teachers found that teachers readily accept AI narrative feedback for formative purposes but insist on human control over summative scoring. Exactly the distinction the HITL architecture preserves.

The HITL model intersects with the broader literature on adaptive learning technology, which encompasses systems that adjust instructional content, difficulty, or pace in response to student performance data. Schaaf et al. (2026) conducted a large-scale empirical study examining the relationship between frequency of intelligent tutoring system (ITS) use and learning gains in secondary mathematics and found a nuanced pattern: ITS frequency of use alone did not significantly predict learning gains, but the combination of ITS use with teacher-structured integration into classroom routines did. This finding has direct implications for the design of the AI Grader implementation scenarios: the technology cannot be expected to produce learning gains simply by virtue of its deployment. AI-generated feedback must be embedded in teacher-mediated instructional structures. This ensures that the feedback is connected to subsequent learning activities. Students also need to be supported in understanding and acting on the feedback they receive. At the same time, teachers must actively use AI-generated data to adapt their instruction.

2.9 Conclusion: Pedagogical Justification for the AI Grader Framework

The analytical review presented in this chapter establishes the theoretical foundations upon which the AI Grader system is built and provides the pedagogical justification for the three implementation scenarios developed in this thesis. The review demonstrates that each component of the proposed AI-assisted assessment approach is grounded in well-established, empirically supported educational theory rather than in technological novelty alone.

The feedback gap identified by Arnold (2011), Black and Wiliam (1998), and Wisniewski et al. (2020) is a genuine and consequential problem in mathematics education: one that is structural in origin and that cannot be addressed by individual teacher effort alone. LLM-based assessment systems, as demonstrated by Biibosunov (2025), Chen and Wan (2025), and Liu et al. (2024), have achieved the technical capability to provide feedback at a quality and scale that was previously impossible. Téllez et al. (2024) nuance this picture, finding that while LLMs produce reliable formative feedback on standard problem types, teacher-supervised verification remains necessary for non-routine problems, directly supporting the HITL oversight built into the AI Grader design. The question this thesis addresses is how to translate that technical capability into pedagogically principled practice. The answer is provided by the theoretical synthesis developed in Sections 1.1-1.9.

Formative assessment theory, as articulated by Black and Wiliam (1998), specifies the pedagogical function that AI feedback must serve: not evaluation but learning support, not judgment but information. ZPD and scaffolding theory, drawing on Vygotsky (1978) and Wood et al. (1976), specify the cognitive mechanism through which that support should operate: calibrated to the individual learner's zone of productive challenge, structured to bridge the gap rather than bypass it. Personalized learning and differentiated instruction theory, as developed by Patrick et al. (2013) and Tomlinson (2017), specify the scale at which support must be individualized to be effective: not at the class level or the group level but at the level of each student's specific reasoning patterns. Self-regulation theory, in the tradition of Zimmerman (2002), specifies the developmental goal that feedback should serve over time: not dependency on external assessment but growing capacity for independent self-monitoring. The human-in-the-loop model, as advanced by UNESCO (2023), Schaaf et al. (2026), and OECD (2026), specifies the governance structure that keeps all of these principles aligned with the educational purposes that teachers and institutions define.

Together, these frameworks constitute what this thesis terms the Pedagogical Justification Framework for AI-Assisted Mathematics Assessment. This framework does not merely describe the theoretical landscape surrounding AI in education. It also provides actionable design criteria for AI Grader implementation. The three implementation scenarios developed in Chapter 3: exit-ticket formative assessment, daily homework feedback automation, and differentiated instruction

support. Each of them is designed to operationalize specific combinations of these theoretical principles, and the teacher interview methodology described in Chapter 3 is designed to evaluate whether the scenarios succeed in translating theory into effective, sustainable classroom practice.

The chapter thus provides both the intellectual scaffolding for the empirical research that follows and the theoretical lens through which its results will be interpreted. The analysis of teacher interview data in Chapter 4 serves as an empirical test of the integrated framework articulated here. It examines whether the framework corresponds to the lived pedagogical experience of experienced mathematics teachers. This test will either confirm the framework's practical validity or reveal the ways in which theory and practice remain to be reconciled.

3 Methodology

3.1 Implementation scenarios for the AI Grader in school

3.1.1 *Scenario 1: End-of-Lesson Exit Tickets for Immediate Formative Assessment*

A mathematics teacher uses the final 5 minutes of each lesson to assess whether students have grasped the lesson's key concepts. Students photograph their handwritten responses to 1–2 targeted problems and submit via Google Forms. The AI Grader processes all submissions and, within several hours after class, the teacher receives a dashboard categorizing students into three groups:

- "Got It": students who demonstrated clear understanding;
- "Almost Got It": students with partial understanding and specific misconceptions;
- "Not Yet": students who struggled with fundamental concepts.

Students simultaneously receive personalized, process-level feedback via email.

Teacher Actions: Upon reviewing the dashboard, the teacher makes a lesson-planning decision calibrated to the distribution of results. The default thresholds work as follows: if a high proportion of students (e.g., around 75–85%) fall in the "Got It" category, new content proceeds as planned; if a moderate proportion (e.g., roughly half to three-quarters) achieved mastery, a short targeted re-teaching mini-lesson (typically 10–15 minutes) opens the following class, addressing the most common misconception identified by the AI; if fewer than half the class demonstrated understanding, a longer block (e.g., 15–25 minutes) is dedicated to comprehensive re-teaching.

Critically, the teacher treats these thresholds as default heuristics, not rigid rules. For example, a single foundational misconception shared by even a small subset of students (e.g., 20–30%) may warrant re-teaching even when the overall "Got It" rate is high. The teacher reviews a sample of AI-generated feedback comments before the next class to verify accuracy and add personal annotations where needed, in line with the human-in-the-loop principle.

At the start of the following lesson, students categorized as "Not Yet" spend a brief opening window (e.g., 5–10 minutes) attempting a parallel redo problem (same concept, different numbers) generated by the AI. This activates the self-reflection phase of Zimmerman's self-regulation cycle and ensures that feedback is acted upon rather than passively received.

Pedagogical Justification: This scenario directly operationalizes formative assessment theory by turning brief exit tasks into high-frequency, low-stakes checks of understanding that immediately shape instruction. The redo mechanism ensures feedback functions as a bridge across the ZPD rather than a terminal evaluation. The teacher's role as interpreter and decision-maker, rather than passive recipient of AI output, reflects the HITL model's requirement that AI-generated data inform but not replace professional pedagogical judgment.

3.1.2 Scenario 2: Daily Homework Review and Feedback Automation

A mathematics teacher manages a large cohort of students (> 100) across several classes (≥ 4), each receiving daily homework assignments. Students submit photographed homework via Google Forms and, within approximately one hour of submission, the AI Grader processes their work and sends personalized feedback that identifies specific errors, awards partial credit for correct steps, and suggests concrete next actions. The feedback is calibrated to the student's demonstrated level: students with surface errors receive process-level explanations ("your sign error occurs at step 3 when moving the term across the equals sign"); students with deeper misconceptions receive scaffolded guidance including a worked example of a closely related problem and a guiding question to prompt self-correction. Students are also prompted with brief self-reflection questions (for example: "What was your reasoning at this step? What would you do differently?"). This activates the self-reflection phase of Zimmerman's SRL cycle Zimmerman (2002).

Teacher Actions: Before class, the teacher reviews the class-wide error patterns dashboard. For example, "a significant cluster of students made sign errors when solving linear equations; several others showed no understanding of the balancing principle." The teacher uses this data to make three decisions: (1) whether to open the lesson with a whole-class discussion targeting the most common error pattern; (2) whether any individual student's feedback requires a personal override or addition before it is seen by the student; and (3) whether the homework task design itself needs adjustment: if the same misconception recurs

across multiple assignments, the problem set may be poorly scaffolded.

The teacher is not expected to read every individual submission, but reviews a small sample of flagged cases (e.g., 5–15) that the AI Grader has identified as ambiguous or borderline. This targeted review preserves teacher authority while significantly reducing grading time (estimates typically range from 70–85% reduction, depending on class size and task complexity).

Pedagogical Justification: This scenario addresses the feedback gap by providing timely, specific, process-oriented feedback that traditional manual grading cannot achieve at scale. By differentiating the depth and tone of feedback by student level, scaffolded for struggling students and Socratic for advanced ones, the scenario operationalizes ZPD and scaffolding fading principles. The self-reflection prompt transforms homework from a passive submission into an active self-monitoring exercise, directly developing the metacognitive habits that Wisniewski et al. (2020) identify as producing the highest-impact learning gains.

3.1.3 Scenario 3: Differentiated Instruction Support System

A mathematics class has students at widely varying levels. Some struggle with basic fraction operations while others are ready for algebraic thinking. The scenario begins with a short diagnostic pre-task: at the start of a new unit, students complete 3–4 carefully designed problems that span the full range of prerequisite knowledge. The AI Grader evaluates these submissions and automatically assigns students to one of three readiness tiers, generating an initial ZPD map for the class. The teacher reviews and adjusts tier assignments where professional judgment warrants. For instance, re-assigning a student whose performance on the diagnostic does not match prior observed work. Tier labels are never shared with students; they are a planning tool for the teacher, not a visible ability classification.

Based on this ZPD mapping, the teacher assigns tiered homework problems matched to each student's readiness level. After submission, the AI Grader evaluates work and generates differentiated follow-up recommendations:

- Mastery students → extension challenges that introduce the next layer of complexity;
- Partial understanding students → scaffolded practice problems with worked examples and guiding questions;

- Foundational misconception students → targeted mini-lesson content addressing the specific gap, with step-by-step re-teaching.

Tier assignments are dynamic: after each submission cycle, the AI Grader updates each student's readiness profile. A student who completes scaffolded problems successfully is automatically recommended for promotion to the next tier at the teacher's discretion.

Teacher Actions: The teacher's primary role in this scenario shifts from task-creator to curriculum architect and monitor. Before the unit begins, the teacher designs or approves the three tiered task sets. The AI Grader can generate draft variants of the same problem at different complexity levels, which the teacher reviews and refines. During the unit, the teacher monitors the tier-movement dashboard weekly: students who are not progressing within their tier after two cycles are flagged for a brief one-to-one conversation. The teacher also watches for students consistently placed in the lowest tier. This pattern may indicate that the misconception is deeper than the AI's feedback can address alone and requires direct human intervention. In whole-class instruction, the teacher plans activities, such as peer explanation tasks or whole-class discussion of a common misconception, that cut across tiers and build the collaborative dimension of learning that purely individualized AI feedback cannot provide, consistent with Vygotsky's emphasis on peer interaction as a driver of ZPD development.

Pedagogical Justification: This scenario operationalizes personalized learning and differentiated instruction by tailoring both tasks and feedback to individual readiness levels. The dynamic tier system and teacher oversight realize Tomlinson's principle that differentiation must be flexibly grouped i.e. students move between tiers as competence grows, preventing tier labels from hardening into fixed ability categories. ZPD and scaffolding guide both the design of tiered tasks and the AI-generated support, ensuring each student experiences appropriate challenge. The teacher's role as curriculum architect and human override preserves professional authority in line with the HITL model, while the AI handles the logistically demanding work of diagnosing readiness and generating differentiated feedback at scale.

3.2 Technical architecture of the AI Grader prototype

This thesis proposes a pedagogically grounded framework for LLM-supported feedback in secondary mathematics. A working software prototype makes that framework concrete. This study refers to it as the **AI Grader** (*Aisuluu*). This section explains what the system does in software. It explains how data move through the pipeline. It also shows where human oversight stays possible in the design. The goal is not full technical documentation. The goal is to specify the architecture that carries out the scenarios in Section~3.1.

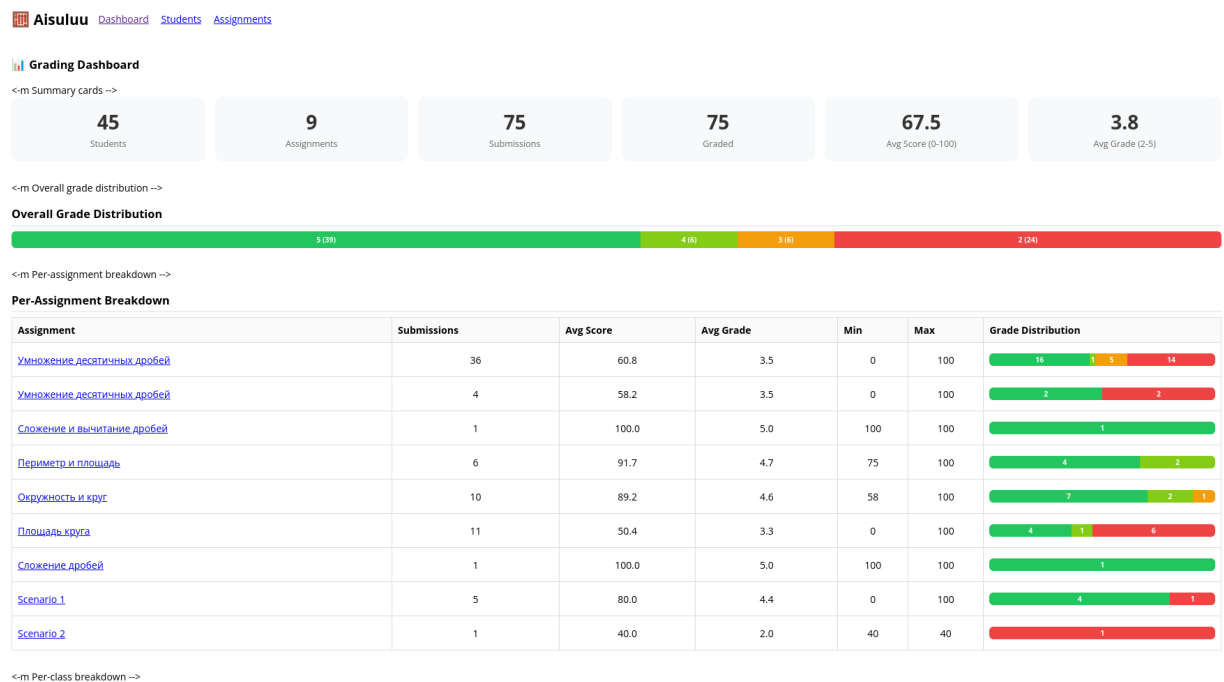


Figure 3.1: Main dashboard of the AI Grader prototype. The interface displays class summaries, assignment tracking, and submission statistics that support teacher monitoring and decision-making within the human-in-the-loop assessment model.


[Aisuluu](#)
[Dashboard](#)
[Students](#)
[Assignments](#)

Assignments

Create New Assignment

Title:

Description:

[Create Assignment](#)

Existing Assignments

ID	Title	Description	Google Form	Created	Actions
1	Умножение десятичных дробей	—	—	2026-02-23 14:00	Edit View
2	Умножение десятичных дробей	—	—	2026-02-26 11:48	Edit View
3	Сложение и вычитание дробей	—	—	2026-03-11 14:53	Edit View
5	Периметр и площадь	—	—	2026-03-11 16:18	Edit View
6	Окружность и круг	ДЗ	Student form Edit form	2026-03-14 10:56	Edit View
7	Площадь круга	ДЗ	Student form Edit form	2026-03-16 14:04	Edit View
8	Сложение дробей	Сложение дробей	Student form Edit form	2026-03-25 11:11	Edit View
9	Scenario 1	Scenario 1	Student form Edit form	2026-04-21 04:40	Edit View
10	Scenario 2	Scenario 2	Student form Edit form	2026-04-25 11:27	Edit View

Figure 3.2: Assignments dashboard of the AI Grader prototype. The interface allows teachers to create assignments, manage linked Google Form submissions, and organize assessment workflows within the automated feedback pipeline.

3.2.1 Design scope

The prototype is designed around the workflow described in the use-case scenarios. Students upload photographs of their handwritten solutions, which are processed and converted into digital mathematical expressions. The system then compares the submitted work with a teacher-defined reference solution, generates a score, and provides structured, process-oriented feedback. Results are stored for teacher review, while feedback is delivered to students via e-mail. The prototype follows an **assignment-centric** approach, where each assignment is defined by a teacher-created reference solution together with relevant metadata.

3.2.2 High-level architecture

The system follows a linear pipeline with a small set of well-defined stages:

1. Ingestion: fetch structured responses from a Google Form, including links to image files stored in Google Drive.
2. Acquisition: download submission images to storage.

3. Text extraction (OCR): convert handwritten mathematics in images into normalized text using multimodal large language models accessed through a unified API.
4. Grading: compare the extracted student work to the reference solution using a text-focused model, with optional multimodal cross-check when original images are passed to the grader so that OCR errors are less likely to dominate the judgment.
5. Persistence: write scores, feedback, and trace fields to a relational database.
6. Notification: send feedback to students via SMTP e-mail.
7. Teacher-facing review: web dashboard (FastAPI) for browsing classes, assignments, and aggregated results.

Orchestration is implemented in Python. Persistent state uses SQLite via SQLAlchemy.

3.2.3 Teacher-authored assignments and rubric-aware parsing

Each assignment contains a reference solution written as text. The prototype supports two formats: a simple legacy format with only plain solution text, and a structured format that can include optional YAML front matter.

The structured format can store additional information such as grading rubrics for each question, scenario identifiers for differentiated tasks or exit tickets, links to learning resources, common error categories, and hints for teacher dashboards.

When an assignment is processed, the reference solution is parsed once and used in several ways. First, it creates an OCR context string that helps vision models recognize the expected mathematical notation and task structure. Second, rubric information can be added to the grading prompt. Third, the parsed data is stored as a structured object for dashboards and remedial learning resources.

This design is pedagogically important because teachers define the grading criteria and resources themselves. The system only uses this information to guide model behavior rather than generating rubrics automatically.

```

total_points: 5
grading_instructions: "Реши сам эти задания столбиком. Я не буду давать тебе решения. Студенты должны решить все задания столбиком. Если будет только ответ без решения, то ставь 0 баллов. ВАЖНО:
проверяй только финальный ответ (число под последней чертой). Если финальный ответ совпадает с правильным ответом для задачи – ставь полный балл, даже если ученик записал условие задачи с другими
операндами. Ошибка в записи условия – это, скорее всего, ошибка OCR-распознавания, а не ошибка ученика."
partial_credit: 0.5
resource_threshold: 80
resources:
  - title: "Нужно повторить умножение столбиком"
    url: "https://youtu.be/juLRpqv0B0E7s1=cpjvV4D0f9GTTbIt"
  - title: "Нужно повторить умножение десятичных дробей"
    url: "https://youtu.be/wq4qY80A6Hg7s1=kmZPw6EV9tpkVu6"
exercises:
  - description: "Решите столбиком эти примеры 4,2 * 3 "
  - description: "Решите столбиком эти примеры 6,7 * 0.3 "
  - description: "Решите столбиком эти примеры 0.22 * 0.06 "
questions:
  - id: 1
    section: "Секция A"
    max_points: 1
  - id: 2
    section: "Секция A"
    max_points: 1
  - id: 3
    section: "Секция A"
    max_points: 1
  - id: 4
    section: "Секция A"
    max_points: 1
  - id: 5
    section: "Секция A"
    max_points: 1
  ---
# Секция A
1. 5,8 * 3
2. 7,9 * 0,6
3. 0,25 * 0,04
4. 0,125 * 0,5
5. 37,5 * 4

```

Figure 3.3: Example of a structured reference file for the AI Grader prototype. The YAML block stores assignment metadata: total points, grading instructions for the language model, partial credit settings, a resource threshold, linked video resources for remediation, practice exercises, and per-question caps in Section A. The lines below the front-matter give the task text shown to students. The grading instructions encode teacher policy, including emphasis on final answers and tolerance for OCR noise in problem statements.

3.2.4 OCR Subsystem

Handwritten mathematics is difficult to recognize because student work may contain poor lighting, tilted images, crossed-out answers, and different notation styles. These problems often reduce the accuracy of traditional OCR systems.

To address this, the prototype uses LLM-based vision models for transcription. The system configuration allows different vision models to be selected without changing the code, and model settings such as temperature can also be adjusted.

The prototype can also use a multi-stage OCR pipeline. For example, several OCR passes may be performed on the same image, while a faster text model can combine or normalize the results. An additional verification step may re-check important parts of the work. These stages improve accuracy but increase API costs.

Before OCR processing, the system may apply image preprocessing methods such as correcting orientation or improving contrast. This helps make handwritten text easier to recognize.

The extracted text is treated as supporting evidence rather than absolute truth. When original images are available, the grading process can still refer back to them directly.

3.2.5 Grading Subsystem

The grading process evaluates the assignment as a whole. The model receives the assignment title, the full reference solution, optional rubric information, OCR text, and, when available, images of the student's work.

The prompts instruct the model to return structured output. This usually includes a score from 0 to 100, written feedback, lists of strengths and areas for improvement, and detailed comments for teacher review.

Whenever possible, responses are generated in JSON format because this makes the data easier to parse and store. If the JSON format is incorrect, the system can retry with stricter instructions.

To reduce formatting errors, mathematical expressions in feedback use simpler formats such as Typst-style inline math instead of complex $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ notation with many escape characters.

The system can also convert the 0–100 score into a smaller grade scale, such as five grade levels. The score thresholds are configurable to match local grading systems, while the detailed numeric scores are still stored in the database.

When student images are included, the model can compare the OCR text with the actual handwritten work. This improves the reliability of partial credit grading and helps identify student misconceptions more accurately.

Overall, the system supports a human-in-the-loop approach. Automation improves speed and consistency, while teachers maintain control through their own reference solutions, grading criteria, and review process.

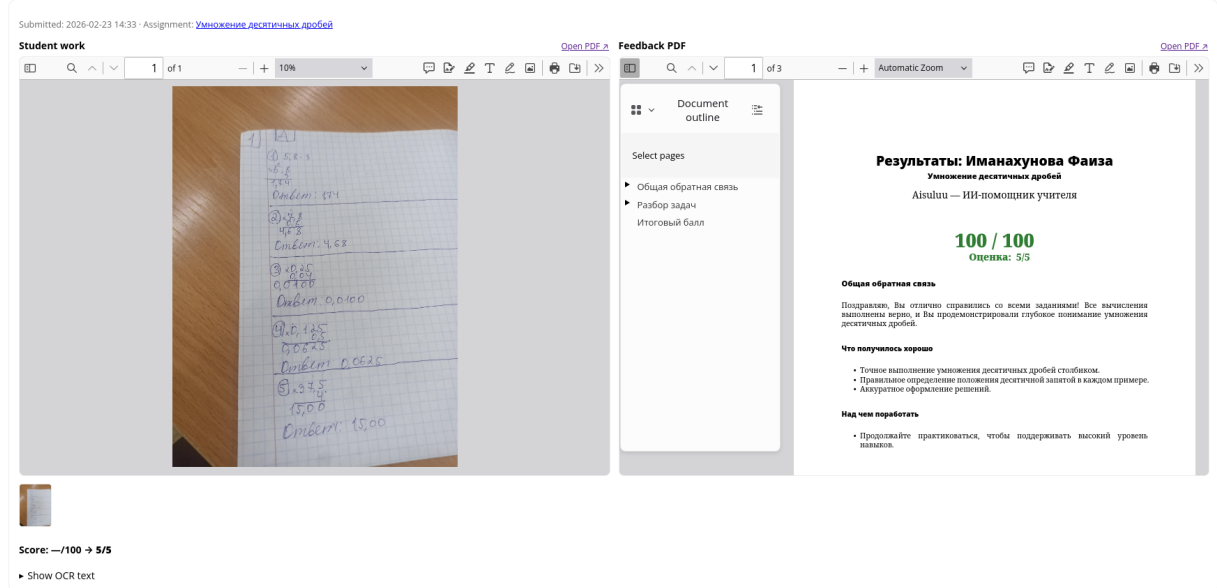


Figure 3.4: Teacher-facing review screen of the AI Grader prototype (AIsuluu dashboard). The left pane shows the student's photographed handwritten work. The right pane shows the generated feedback PDF with summary score, narrative feedback, and section outline. The layout supports comparing visual evidence with model output. The side-by-side layout enables the teacher to verify OCR accuracy and override AI-generated scores directly from this screen, fulfilling the human-in-the-loop checkpoint.

Illustrative materials. This section describes the system's architecture in prose. The figures above show the teacher dashboard, assignments management, a structured reference solution, and the teacher review screen. These visuals help illustrate how the prototype developed over time. Appendices A and B provide PDF examples of graded student work. These examples are arranged to show gradual improvements in OCR accuracy, grading structure, and the quality of feedback.

3.3 Focus group

The empirical component of this work was conducted through a focus group with practicing secondary school mathematics teachers. This method was selected because the research objective at this stage was not to measure outcomes but to elicit rich, experience-based professional judgment about the three AI Grader

implementation scenarios developed in Section~3.1. A focus group format was particularly appropriate given that the scenarios involve shared professional contexts: grading workload, classroom feedback practices, and technology integration, where collective discussion among peers produces richer, more contextually nuanced data than individual responses alone. Morgan (1997) establishes the foundational rationale for this choice. He argues that focus groups are particularly well-suited to research on shared professional practices. Group interaction surfaces the norms and assumptions that structure individual behavior in ways that one-to-one interviews cannot replicate. Moreover, this methodological choice is well supported in the literature: Gil-Flores and Alonso (1995) demonstrated that focus groups generate contextually nuanced data precisely because participants challenge each other's professional reasoning, while Kitzinger (1995) established that peer interaction surfaces assumptions that individual interviews leave implicit. Participants could share experiences, question each other's thinking, and reach conclusions through open discussion, which fits the qualitative nature of this research phase.

3.3.1 Rationale for a Qualitative Design

The choice of qualitative methodology reflects the nature of the research questions addressed in this phase. The study does not seek to measure whether the AI Grader produces statistically significant learning gains. This would require a controlled experimental design beyond the scope of this thesis. Instead, it seeks to assess whether the proposed implementation scenarios are pedagogically coherent and practically viable from the perspective of the teachers who would use them. This requires access to professional judgment, contextual knowledge of local school conditions, and articulated reasoning about trust, control, and classroom feasibility. All of which are best captured through open, exploratory qualitative inquiry rather than structured quantitative instruments. Creswell and Poth (2018) describe this as the core epistemological commitment of qualitative inquiry. When the aim is to understand meaning, process, and contextual conditions rather than to measure outcomes, qualitative methods are not merely appropriate. They are methodologically necessary.

3.3.2 Participant Selection

Six secondary school mathematics teachers participated in the focus group. Teachers were selected on the basis of direct relevance to the research context:

all six were actively teaching mathematics at the secondary level (grades 5–9) in Kyrgyzstan, and had firsthand experience with the practices under investigation. Specifically, the grading of student work and the provision of written or oral feedback. No minimum or maximum years of experience was imposed as a selection criterion; the priority was to recruit teachers currently engaged in the day-to-day realities of mathematics assessment rather than teachers whose experience was primarily administrative or historical. All teachers consented to audio recording of the session and to the use of their responses in anonymized form for research purposes. No names, school affiliations, or identifying details appear in the analysis; teachers are referred to throughout this thesis by neutral labels (e.g. Participant 1 through Participant 6).

3.3.3 Focus Group Protocol and Structure

The focus group was conducted in Russian and Kyrgyz. These were the shared professional and instructional languages of all participants. Conducting the session in these languages ensured that responses were not limited by language barriers. It also allowed participants to express nuanced professional views naturally. The session lasted approximately 95 minutes and was audio-recorded with participants' consent.

The protocol was structured across seven thematic sections. These sections moved from general teaching context toward increasingly specific discussion of the AI Grader scenarios. The opening section established participant background, including teaching load, grade levels, and approximate number of students per week. This contextual grounding was essential for interpreting the responses that followed. The practical significance of a feedback automation system differs substantially between a teacher managing two small classes and one managing five sections of thirty students each.

The second and third sections focused on current feedback and formative assessment practices. Participants discussed what feedback they currently provide, what constraints prevent more individualized or timely feedback, and how they use assessment data to differentiate instruction. These sections were designed to reveal the existing gap between desired and actual feedback practice. This is the same gap that Chapter 2 identifies as the structural driver of the AI Grader's pedagogical justification. By establishing this baseline, the protocol ensured that teacher evaluations of the AI scenarios were grounded in genuine professional need

rather than abstract speculation.

The fourth section assessed participants' existing familiarity with digital tools and AI. This included platforms such as Google Forms, electronic grade books, and large language model tools such as ChatGPT. This section served to calibrate the interpretive context. A teacher who has never encountered automated assessment will evaluate the scenarios differently from one who has experimented with AI-assisted tools. Both types of responses carry analytical value.

The fifth and central section presented the three implementation scenarios from Section 3.1 in sequence. These were the exit-ticket formative assessment scenario, the daily homework feedback automation scenario, and the differentiated instruction support scenario. Participants were invited to evaluate each scenario in a structured way. For each one, they assessed its perceived utility, its practical feasibility within their school context, their level of trust in AI-generated feedback, and how the scenario would affect their professional role. Participants were also asked to compare the three scenarios and identify which they would prioritize and why. This comparative discussion was designed to surface not only individual preferences but also the underlying criteria through which practicing teachers evaluate instructional technology. These criteria included time efficiency, feedback quality, lesson planning support, and student motivation.

The sixth section addressed risks, concerns, and implementation conditions. Participants discussed what barriers could prevent adoption, what safeguards would be required, and what specific features would need to be in place for the system to work in Kyrgyzstani school conditions. These features included language support, alignment with local curriculum standards, technical infrastructure, and teacher training. This section was included because the theoretical framework in Chapter 2 identifies human-in-the-loop oversight and local contextual adaptation as essential conditions for responsible AI integration in education. Teacher responses in this section provided empirical grounding for those theoretical claims.

The final section invited open-ended reflection: each participant was asked to summarize the role they believed AI could appropriately play in mathematics education, and to raise any dimensions of the issue that the structured questions had not captured.

Table 3.1 maps each section of the focus group protocol to the corresponding research objectives and theoretical themes established in Chapter 2, demonstrating

that the data collection instrument systematically addresses all dimensions of the study’s analytical framework.

Table 3.1: Alignment of Focus Group Sections and Questions with Research Objectives and Theoretical Framework

Protocol Section	Questions	Research Objective	Theoretical Theme (Ch. 2)
Section 1: Teaching Context	Q1–Q2	RO4: Validate proposed scenarios through teacher perspectives	Feedback Gap (Black & Wiliam, 1998); Formative Assessment
Section 2: Current Feedback Practice	Q3–Q5	RO4: Validate proposed scenarios; RO3: Design implementation scenarios	Feedback Gap; Formative Assessment (Leahy et al., 2005)
Section 3: Formative Assessment & Differentiation	Q6–Q8	RO3: Design implementation scenarios	Differentiated Instruction (Tomlinson, 2017); Formative Assessment
Section 4: Technology & AI Familiarity	Q9–Q10	RO4: Validate proposed scenarios	Human-in-the-Loop Model (Kasneci et al., 2023)
Section 5: AI Grader Scenario Evaluation	Q11–Q14	RO3: Design scenarios; RO4: Validate scenarios	ZPD & Scaffolding (Vygotsky, 1978; Wood et al., 1976); Differentiated Instruction; Formative Assessment
Section 6: Concerns, Control & Conditions	Q15–Q18	RO4: Validate proposed scenarios	Human-in-the-Loop Model; Personalized Learning (Patrick et al., 2013)
Section 7: Closing Reflection	Q19–Q20	RO4: Validate proposed scenarios	Pedagogical Justification Framework (all themes integrated)

3.3.4 Data Collection Procedure

The focus group was conducted in a conversational register, without technical jargon, and the moderator followed the protocol’s guidelines for creating an environment in which participants felt free to agree, disagree, and build on each other’s contributions. The moderator did not present the AI Grader scenarios as existing or already-deployed systems, but explicitly framed them as proposed designs being evaluated for practical validity. This framing was intentional: it positioned teachers

as expert evaluators rather than passive recipients of a technology decision, which is consistent with the human-in-the-loop philosophy that underpins the entire AI Grader framework.

The full session was audio-recorded, and the recording was subsequently transcribed. Responses were anonymized at the transcription stage, with all references to participant names, schools, and other identifying details removed before analysis.

3.3.5 Analytical Framework

The focus group data were analyzed using a combined approach of thematic analysis and content analysis, applied sequentially to the transcribed material.

Thematic analysis followed the six-phase procedure described Braun and Clarke (2006, 2022) and was conducted in the reflexive tradition elaborated in their subsequent work in which researcher engagement with the theoretical framework is understood as an analytical resource rather than a source of bias. Kushnir (2025) provide specific methodological guidance for applying thematic analysis in educational research contexts, confirming the suitability of the reflexive approach for capturing practitioner reasoning and professional judgment. The first phase involved familiarization with the data through repeated reading of the transcript. The second phase focused on generating initial codes that captured meaningful units of teacher response. In the third phase, codes were organized into candidate themes. The fourth phase involved reviewing and refining those themes against the full dataset. The fifth phase defined and named the final themes. The sixth phase produced the analytic narrative.

The thematic analysis was conducted inductively in the first instance. This allowed themes to emerge directly from the data. The themes were then interpreted through the deductive lens of the theoretical framework established in Chapter 2. This meant that emergent themes were examined for how they corresponded with, or diverged from, key pedagogical constructs. These constructs included formative assessment, ZPD and scaffolding, differentiated instruction, personalized learning, and the human-in-the-loop model. Where teacher responses confirmed the practical validity of a theoretical construct, this was noted as convergent evidence. Where responses identified gaps, contradictions, or conditions not anticipated by the theoretical framework, these were treated as refinements to the AI Grader design.

Content analysis was applied as a complementary method. It was applied following the directed approach described Hsieh and Shannon (2005) and Mayring (2000), in which coding categories are derived from an existing theoretical framework rather than generated purely inductively from the data, and conducted within the broader qualitative tradition of content analysis outlined Elo and Kyngäs (2008). Its purpose was to systematically identify the frequency and distribution of key evaluative categories across the transcript. Each substantive participant turn was coded according to seven dimensions derived from the focus group protocol's own analytical framework. These dimensions were: (1) current feedback practices and their limitations; (2) perceived utility of AI-generated feedback; (3) perceived feasibility within local school conditions; (4) trust in AI accuracy; (5) concerns about student dependency or reduced teacher agency; (6) required implementation conditions; and (7) references to specific contextual factors in local schools, including language of instruction, internet and device access, and curriculum alignment. This structured coding allowed for a systematic cross-participant comparison of response patterns across all three scenarios. It also provided a degree of analytical rigour that purely interpretive thematic analysis alone would not have achieved. Elo and Kyngäs (2008) provide complementary methodological guidance for the content analysis process in educational research. They specify criteria for rigorous category construction. These criteria were applied here to ensure that coded units remained anchored in the transcript data rather than in the researcher's prior theoretical assumptions.

The two methods were used in an integrated fashion. Thematic analysis provided depth and interpretive richness. It captured the reasoning and professional logic behind teacher positions. Content analysis provided breadth and comparability. It identified which concerns and endorsements were shared across participants and which were individual. Together, they produced an analytical account of the focus group data that is both contextually sensitive and systematically grounded.

3.3.6 Limitations

Several methodological limitations of this phase of the research should be acknowledged. First, the sample of six participants, while appropriate for a qualitative focus group and sufficient to generate substantive discussion, does not permit generalization to the broader population of secondary mathematics teachers in Kyrgyzstan or elsewhere. The findings represent the perspectives of a specific

group of practitioners in a specific institutional context, and should be interpreted as indicative rather than representative. Second, the focus group format, while productive for eliciting collective professional reasoning, may have introduced dynamics of social desirability or peer conformity that suppressed dissenting positions. The moderator's explicit encouragement of disagreement was designed to mitigate this risk, but it cannot be fully eliminated. Third, the scenarios were presented in described rather than enacted form: participants evaluated hypothetical implementations rather than responding to lived experience of the system. Their assessments of feasibility, trust, and utility are therefore prospective judgments rather than retrospective evaluations, which introduces a degree of uncertainty about whether actual deployment would produce the same responses. These limitations do not invalidate the findings but appropriately bound their scope and inform the interpretation of results in Chapter 4.

4 Results

This chapter presents the empirical findings from the focus group conducted with six secondary school mathematics teachers in Kyrgyzstan. Results are organized in two parts: directed content analysis (Section~4.2) followed by reflexive thematic analysis (Section~4.3), together addressing Research Objective 4 (validation of the proposed AI Grader scenarios).

4.1 Overview of the Focus Group Data

The empirical phase of the study consisted of a single focus group with six mathematics teachers from secondary schools in Kyrgyzstan, whose teaching experience ranged from first-year teachers to 10 years of practice. The teachers collectively taught between 33 and 170 students each, across grades 5 to 11, and represented a mixture of primary background, subject specialists, and ORT preparation tutors. The group discussion lasted approximately ninety-five minutes and was conducted in a mixture of Russian and Kyrgyz in a conversational register without technical jargon, in line with the protocol described in Chapter 3.

The focus group generated a rich corpus of data on current feedback practices, experienced workload, perceived opportunities and risks of AI-assisted assessment, and contextual conditions for implementation in Kyrgyz schools. The session was audio-recorded and transcribed verbatim, with all personal and institutional identifiers removed at the transcription stage. The analysis that follows presents, first, the results of the directed content analysis, and second, the themes that emerged from the reflexive thematic analysis.

4.2 Content Analysis

4.2.1 Coding Framework and Procedure

Directed content analysis was conducted using the seven evaluative categories defined in Section~3.3.5: (1) current feedback practices, (2) perceived utility of AI, (3) feasibility, (4) trust in AI, (5) concerns about dependency or reduced teacher agency, (6) implementation conditions, and (7) contextual factors specific to local schools. Each substantive participant turn in the transcript was coded against one or more of these categories, resulting in 52 coded units across the entire dataset.

Table 4.1 summarizes the distribution of coded units across the seven categories and indicates which participants contributed most heavily to each dimension.

Table 4.1: Distribution of Coded Units Across Content Analysis Categories

Category	Coded Units	Salient Contributors
Current feedback practices	9	All six teachers
Perceived utility of AI	10	Participant 6, Participant 5, Participant 1
Feasibility	7	Participant 6, Participant 4, Participant 5
Trust in AI	6	Participant 5, Participant 6, Participant 1
Concerns about dependency / reduced teacher agency	8	Participant 5, Participant 6, Participant 3
Implementation conditions	7	Participant 6, Participant 5, Participant 1
Contextual factors (local schools)	5	Multiple participants
Total	52	

Two participants, #5 and #6, accounted for over half of all coded units, reflecting their high level of engagement in the discussion. At the same time, even the less vocal participants contributed detailed accounts on specific issues such as workload and classroom constraints.

4.2.2 Current Feedback Practices

All six teachers described feedback as predominantly manual, notebook-based, and heavily constrained by time. Participant 2, who teaches four fifth-grade classes, reported collecting student notebooks daily or every second day in line with school standards and spending “2–3 hours every day, four stacks” on checking homework. Participant 4 similarly characterized her workload by stating that “half of the day passes with those notebooks” when teaching primary classes with multiple exercise books per subject.

The strategies used to convey feedback varied. Some teachers, such as Participant 2, write out full correct solutions in red pen and then mark answers as correct or incorrect, while others, such as Participant 5, deliberately avoid providing

worked solutions and instead underline only the place where the error occurs so that the student must analyze the mistake. Several teachers reported using quick in-class tests as a more manageable way to monitor understanding and generate grades than continuous notebook checking. Participant 3 acknowledged that checking homework only once every two weeks is pedagogically suboptimal but described it as the only sustainable rhythm under current conditions.

These accounts empirically confirm the “feedback gap” described in the theoretical chapter: even motivated teachers cannot provide frequent, high-quality, individualized feedback to every student under the prevailing structural conditions of class size and workload.

4.2.3 Perceived Utility of AI

Ten coded units concerned perceptions of how AI tools could usefully support assessment and feedback. Teachers already using AI in some form (e.g. Notebook LM for ORT preparation, ChatGPT during lessons, or Photomath for personal problem solving) tended to articulate more concrete potential benefits. Participant 5 reported that after an AI assistant successfully generated a mathematical graph she had assumed it could not produce, she “started to believe” in its capabilities, although she still treats its outputs cautiously. Participant 1 uses Photomath and ChatGPT to clarify problems she finds difficult before explaining them to her class, viewing AI as a private planning tool rather than a classroom-facing system.

When discussing the three AI Grader scenarios, all participants considered the third scenario (pre-topic diagnostic assessment) to be the most useful and least risky, particularly because it can provide a “general picture” of the class’s readiness before starting a new topic and thus support planning differentiated instruction. Participant 2 explained: “in my own diagnostics I don’t always reach everyone, here there would be a general picture, and it would already be clear how to structure the lesson going forward”. Scenario 1 (End-of-Lesson Exit Tickets) was viewed as potentially valuable for formative assessment, whereas Scenario 2 (Daily Homework Review) attracted more skepticism regarding both efficiency and robustness.

4.2.4 Feasibility and Implementation Conditions

Feasibility and implementation conditions were discussed together by the participants and are therefore closely related in the data. On the technical side, sev-

eral teachers expressed concern about the ability of any system to reliably recognize handwritten mathematics in both Russian and Kyrgyz and to distinguish between individual students' work when photos are shared or combined. Participant 6 suggested that in real-time classroom settings, structured tools such as Google Forms might be more efficient than photographing notebooks for automated analysis.

On the infrastructural side, participants highlighted uneven device access and network connectivity. While “almost all” of Participant 1's students own smartphones, at least one pupil in her class uses only a basic Nokia phone and cannot access digital materials such as presentations. More significantly, both Participant 4 and Participant 5 emphasized that national or local bans on student phone use during lessons present a structural barrier to any scenario that presupposes in-class device access. At the teacher level, Participant 2 described a colleague who is both a programmer and a mathematician yet refuses to use AI on principle, indicating that professional attitudes to AI cannot be inferred from digital competence alone.

4.2.5 Trust in AI and Concerns about Dependency

The teachers did not treat AI systems as either fully trustworthy or inherently unreliable. Rather, trust was described as something that must be built through repeated verification. Participant 5 outlined this process explicitly: if a teacher cross-checks AI feedback against their own judgment for “one, two, three weeks” and finds it consistently accurate, then “trust will probably develop”. This trust-building model aligns with the HITL orientation of the AI Grader design, where the teacher remains the final arbiter of assessment decisions.

At the same time, participants expressed clear concerns about students becoming dependent on AI or using it as a way to avoid thinking. Participant 1 worried that “children will simply stop thinking” if all checking is delegated to AI and that disputes over grades may be redirected to the system rather than discussed with the teacher. Participant 3 argued that AI should not “chew everything” for students, because ready-made solutions undermine the development of independent problem-solving skills. These concerns directly echo the theoretical emphasis on maintaining productive difficulty within the ZPD and guarding against over-scaffolding.

4.2.6 Contextual Factors in Kyrgyz Schools

Finally, the participants identified several contextual factors specific to Kyrgyzstan that are not fully captured by generic discussions of AI in education. They stressed that any viable system must be at least bilingual or trilingual, as schools across the country teach in Kyrgyz, Russian, Uzbek, Tajik, and Dungan, and that even within Kyrgyz there are significant regional lexical differences that may affect language models. They also emphasized that policy-level constraints, such as phone bans and expectations around notebook checking, shape what is practically implementable in classrooms, regardless of technical capability.

4.3 Thematic Analysis

Six themes were identified through reflexive thematic analysis following the six-phase procedure of Braun and Clarke (2006). The analysis was conducted inductively in the first instance, allowing themes to emerge directly from the data. The themes were then interpreted through the deductive lens of the theoretical framework established in Chapter 2. Two themes achieved universal convergence across all six participants. Two achieved convergence across five of six participants. Two were strongly articulated by three to four participants, with implicit resonance across the rest of the group.

4.3.1 Theme 1: The Feedback Burden: Time, Volume, and Structural Overload

Convergence: all six participants

The most pervasive theme across the focus group is the structural impossibility of providing high-quality, individualized feedback at the scale teachers are expected to operate. All six participants described feedback provision as unsustainable. The volume of students per teacher, combined with institutional requirements for daily notebook checking, creates a workload that none of them can meet without reducing quality or frequency.

The accounts converged not only in their description of the problem but in the specific ways teachers adapt to manage it. These adaptations include reducing the frequency of full notebook checks, substituting individual feedback with whole-class correction, and shortening the scope of written annotation. Participant 3 described accepting a two-week checking delay, even while recognizing that this

makes the feedback pedagogically ineffective. Participant 1's account captures the structural paradox most acutely. Her attempt to provide genuinely individualized feedback to students with prior-knowledge gaps consumed so much instructional time that the class fell a full month behind the syllabus on a single topic. The choice she faced was not between good feedback and no feedback. It was between individualized feedback and curriculum progression. Neither option is educationally adequate.

This theme provides direct empirical confirmation of the feedback gap that Black and Wiliam (1998) identify as endemic to mathematics classrooms. It also corroborates Leahy et al. (2005)'s argument that day-by-day formative assessment is operationally impossible without feedback infrastructure that extends beyond the individual teacher's capacity. The AI Grader's core design rationale addresses precisely this structural problem by automating the time-intensive components of feedback, so that the teacher's professional attention can be directed toward interpretation and intervention.

4.3.2 Theme 2: Heterogeneous Classrooms and the Limits of Uniform Instruction

Convergence: five of six participants

Closely related to the feedback burden, but analytically distinct from it, is the challenge of managing significant within-class variation in student readiness and prior knowledge. All participants except Participant 2 described classrooms in which the range of mathematical competence spans multiple years of prior curriculum content. Uniform instruction is simultaneously under-challenging for stronger students and overwhelming for weaker ones.

Each participant described informal differentiation strategies they deploy in response. Participant 6 prepares six to eight problems and allows students to progress through as many as they can, awarding additional credit for completing more. Participant 5 adjusts the number of tasks assigned to each individual. Participant 3 assigns problems of varying numerical complexity to distribute challenge within the same topic. These strategies share an important characteristic: they are informal and improvised. None constitute the systematic, tiered differentiation that Tomlinson (2017) describes as responsive to individual learner needs across content, process, product, and learning environment. Participant 6 acknowledged this directly: "I can talk about differentiated instruction, but because I have so many

students, it is not always possible".

The theme also foregrounds a diagnostic challenge embedded within differentiation. Before tasks can be tiered, the teacher must know where each student is. Participant 1 identified this as her primary feedback difficulty: "I don't know where the student is stuck... they fell behind, but I don't know exactly where". This diagnostic bottleneck has direct implications for the AI Grader's Scenario 3 design. The pre-lesson tier-mapping function addresses precisely this problem by generating the readiness data that makes systematic differentiation possible, without requiring teachers to administer and evaluate individual assessments manually.

4.3.3 Theme 3: Conditional Openness to AI: Trust Built Through Verification

Convergence: four of six participants with direct AI experience

No participant in the focus group expressed categorical rejection of AI tools. No participant expressed unconditional trust either. Instead, trust emerged from the data as a relational achievement. It is something earned through concrete, verified experiences of accuracy, not assumed on the basis of technical capability.

Participant 5's account of the moment when an AI assistant correctly generated a mathematical graph she had doubted it could produce is analytically significant. It illustrates how teacher trust develops: through specific, confirmable instances of competence in domains the teacher can independently evaluate. She elaborated this into a general principle: sustained self-verification over one to three weeks, in which the teacher checks AI outputs against their own judgment, is the mechanism through which trust can develop. Participant 6 complemented this with a developmental framing. She described piloting as a process that produces "a good product" over time. The expectation is not perfection at initial deployment, but transparent iteration toward reliability.

This theme has a direct implication for AI Grader implementation that extends beyond technical specifications. The pathway to teacher adoption is not a demonstration of capability in the abstract. It is a structured verification phase, in which teachers routinely compare AI-generated feedback with their own assessment before the system is used unsupervised. The HITL architecture already incorporates this principle by design, but the teacher accounts clarify what these checkpoints need to feel like from the teacher's perspective: not a bureaucratic override mechanism, but a genuine opportunity to calibrate trust through iterative practice.

4.3.4 Theme 4: The Irreplaceable Teacher: Emotional Presence and Relational Authority

Convergence: strongly expressed by three participants; implicitly shared across the group

Across all three scenario evaluations, participants drew a consistent boundary between what AI can appropriately perform and what must remain with the teacher. This boundary is not technical. It is relational and emotional. All participants who addressed the question converged on a shared formulation: AI handles cognitive support tasks such as checking work, generating problems, and diagnosing readiness. The teacher retains exclusive responsibility for the emotional, motivational, and relational dimensions of learning.

Participant 4 stated this boundary most precisely: "AI should only explain. Everything else, emotions, the student must receive from the teacher". Participant 6 contributed the focus group's most empirically significant account. In a prior AI-assisted lesson, many students repeatedly refused to continue interacting with the AI system. They wrote: "I don't want to, stop, don't explain, I don't want to". The same students could not disengage from a teacher's direct question in the same way: "a teacher asks them in person and they can't deflect; they have to reason". This asymmetry is not a technical limitation of current AI systems. It is a structural feature of the student-teacher relationship that technology cannot replicate.

Participant 3 added a pedagogical dimension that resonates directly with the scaffolding literature. AI should not "chew everything up" for students, because excessive readiness of support removes the productive cognitive difficulty that learning requires. This echoes Wood et al. (1976) principle of fading: effective scaffolding is temporary and contingent, not maximal and permanent. Together, these accounts validate and enrich the HITL model. Teacher authority is irreplaceable not because AI lacks capability, but because the student-teacher relationship carries motivational and developmental weight that no automated system can substitute.

4.3.5 Theme 5: Scenario Preference: Pre-Lesson Diagnostics as the Entry Point

Convergence: all six participants (unanimous ranking)

When asked to rank the three AI Grader scenarios, all participants placed Scenario 3 (pre-lesson diagnostic assessment) first. This unanimous preference is the

most unambiguous empirical finding of the focus group. It has direct implications for the sequencing of AI Grader deployment.

The reasons participants gave clustered around two complementary advantages. First, Scenario 3 addresses a gap in current practice. As Participant 6 noted, teachers administer informal diagnostic checks but rarely manage to assess all students within the available time. Instructional planning is therefore based on an incomplete picture of class readiness. A system that provides a complete class-wide readiness map before the lesson begins directly addresses this informational deficit. Second, Scenario 3 carries the lowest implementation risk of the three options. It does not involve real-time processing during class. It does not raise concerns about adaptive feedback dependency.

Scenario 1 attracted the most skepticism. Participant 6's core objection was efficiency: "by the time they write it, photograph it, a minute will already pass — that's inefficient. If you open a Google Form right away and during the lesson you monitor what they're solving right there inside, it would be more correct". The additional risks of OCR failure on handwritten mathematics and the possibility of multiple students' work being pooled into a single photograph were also raised. Scenario 2 was valued for its individualization potential, but Participant 5's concern that students may "simply stop thinking" if AI generates all their practice problems was shared by other participants. The scenario ranking — 3, then 2, then 1 — constitutes a de facto recommendation for deployment sequencing. The system should be introduced beginning with the least technically demanding and most immediately transparent scenario, advancing to higher-complexity scenarios once teacher trust has been established through verified experience.

4.3.6 Theme 6: Contextual Barriers: Language, Devices, Policy, and Professional Readiness

Convergence: five of six participants

The final theme brings together a cluster of Kyrgyzstan-specific implementation barriers that the theoretical framework established in Chapter 2 does not fully anticipate. These barriers are not challenges common to all AI-in-education deployments. They are specific to the institutional, linguistic, and policy environment of the schools in which the participants work. They represent refinements to the AI Grader design that emerge from the data rather than from theory.

The most substantive contextual requirement is multilingual functionality.

Participants confirmed that Kyrgyzstan's school system operates across at least five language contexts: Russian, Kyrgyz, Uzbek, Tajik, and Dungan. Regional lexical variation within Kyrgyz adds further complexity. A system that operates only in Russian would be inaccessible to students in a significant proportion of schools. A system that operates only in standard Kyrgyz would be unreliable for students from regions with distinct mathematical vocabulary. This requirement cannot be addressed through pedagogical design alone. It requires language-specific model training or fine-tuning.

The phone-ban policy was confirmed by two participants as actively enforced in their institutions. All scenarios as currently designed require students to photograph and submit work via smartphone during class time. In schools where phones are collected at the start of every lesson, these scenarios are effectively blocked. Device inequality represents a further boundary condition. Participant 1 described a student who uses only a Nokia feature phone and cannot receive digital materials. A fallback pathway is required for equitable access across a full class. Finally, Participant 6's observation about a digitally capable colleague who refuses AI on principle illustrates that professional readiness is not simply a function of technical competence. In a systematic review of technology integration in teacher education, Tondeur et al. (2016) confirms this finding internationally. Attitudinal and pedagogical readiness predict actual classroom technology use more reliably than technical skill. This implies that professional development for the AI Grader must address beliefs and values, not only technical competencies. Any implementation strategy that assumes digital skill is sufficient for adoption will underestimate the attitudinal dimension of teacher buy-in.

5 Discussion

The findings in Chapter 4 address Research Objective 4. This objective is to validate the proposed AI Grader scenarios through the perspectives of practicing mathematics teachers. This chapter interprets those findings in relation to the theoretical framework in Chapter 2.

The discussion follows three guiding questions. First, how far does the empirical data confirm the framework's diagnosis of the feedback problem? Second, how far does the data support the AI Grader scenarios as pedagogically coherent responses to that problem? Third, what refinements to the framework and to the scenario designs are required by findings that the literature review did not anticipate?

5.1 Confirming the Feedback Gap

The focus group data provide strong empirical support for the claim made in Section 2.1. The feedback gap in mathematics education is structural in origin and cannot be closed through individual teacher effort alone. All six participants described feedback as chronically constrained by the ratio of students to available time. They reported delayed checking, reduced annotation, and a shift from individualized comments to whole-class correction. Black and Wiliam (1998) describe exactly these patterns as signs that feedback has been scaled down to remain manageable, not maintained at the level that research identifies as effective.

The data do more than show that a gap exists. They reveal how it appears in the specific context of secondary mathematics in Kyrgyzstan. A mandatory daily notebook-checking policy for lower-secondary classes creates an unusually heavy feedback obligation. It applies to every lesson, not only to occasional tests or assignments. Participant 2's report of spending two to three hours each day on four stacks of notebooks is not an exception. It is a direct consequence of the policy. This institutional detail strengthens the relevance of the AI Grader framework for this context. At the same time, it confirms that the underlying drivers of the feedback gap are the same as those identified in the international literature. The main drivers are class size, time pressure, and workload intensity.

Within-class heterogeneity, described in Theme 2, adds another dimension. The problem is not only how often feedback is given. It is also what level of preci-

sion and calibration the feedback can achieve. Teachers who cannot diagnose each student's current level cannot provide feedback that is aligned with the learner's Zone of Proximal Development, as Vygotsky (1978) defines it. Participant 1's statement that she does not know "where the student is stuck" is a direct expression of this diagnostic deficit. Scenario 3 is designed to produce exactly the formative data that are missing. It provides a class-wide readiness map that enables calibrated feedback and differentiated planning.

5.2 Validating the AI Grader Scenarios

The focus group results provide conditional, but meaningful, validation of all three AI Grader scenarios. The validation is conditional because teachers evaluated descriptions of scenarios, not real deployments. Their judgments are therefore prospective. They anticipate usefulness and risks, rather than reflecting experience after implementation. Even so, the data show that the scenarios address genuine needs. Teachers described concrete ways in which each scenario could support their practice.

Scenario 3 received the clearest support. Its central function is to generate a picture of class readiness before instruction begins. This addresses the diagnostic bottleneck that emerged as a major barrier to differentiated instruction. Participant 6 described Scenario 3 as providing a "general picture" that makes planning possible. This language mirrors the diagnostic dimension of personalized learning described in the theoretical chapter. The unanimous ranking of Scenario 3 as most useful also supports the idea that diagnostic support should be the first stage of AI Grader deployment. Teachers want help to understand where students are before they want automated grading or adaptive task generation.

Scenario 2 received qualified support. Teachers recognized that automated homework review and feedback could relieve a major burden. Producing individualized, process-level feedback for more than 100 students per day is impossible manually. Scenario 2 offers a realistic way to approach this scale. At the same time, Participant 5 and Participant 3 raised serious concerns about student dependency and cognitive effort. They worried that if AI always provides a worked solution or very strong hints, some students may stop thinking actively. This concern fits the scaffolding literature very closely. Wood et al. (1976) argue that scaffolding must fade over time and should not remove all challenge. The risk is that feedback

will bypass the ZPD instead of helping the learner to cross it. The self-reflection prompts in Scenario 2 are a designed response to this risk. They invite students to analyze their own reasoning rather than simply accept an answer. The focus group suggests that these prompts are helpful, but that teacher-led follow-up remains important if students are to act on the feedback.

Scenario 1 received the weakest support. The main objection did not concern AI capability itself. It concerned workload and efficiency. Participant 6 argued that a real-time Google Form already allows teachers to see students answers during the lesson and that taking photos of handwritten work adds extra effort with limited benefit. Technical concerns about handwriting recognition and cheating (for example, several students using the same solution and photo) were also raised. These worries are addressable through improved OCR and clearer classroom routines. The efficiency critique is more fundamental. It suggests that Scenario 1 must clearly show what it offers beyond what simple digital quizzes already provide. The key added value is process-level analysis of working steps, not only final answers. This distinction needs to be made explicit in scenario descriptions and in teacher training. Otherwise, teachers will compare Scenario 1 to tools they already know and conclude that it is not worth the extra friction.

5.3 Refinements to the Framework

The focus group also produced several findings that refine the theoretical framework. These are not contradictions. Instead, they add detail that the literature review did not fully anticipate.

The first refinement relates to the limits of AI engagement. In Chapter 2, the human-in-the-loop model is described mainly as a governance structure. Teachers remain in control. They see AI outputs, edit them, and override them when necessary. This description is accurate. However, the focus group showed an additional layer. Participant 6 reported that many students refused to continue a dialogue with an AI assistant. They wrote messages such as “I do not want to, stop, do not explain”. A student cannot so easily disengage from a teacher who asks a direct question in a classroom. The teacher–student relationship has a compulsory element. AI interaction does not. This asymmetry has important consequences. It suggests that AI feedback, if presented as an optional extra, may simply be ignored by some students. To address this, AI Grader activities should be integrated into

teacher-led routines. For example, the teacher could start the next lesson by asking students to respond to a specific point from their AI feedback in front of the class. In this way, AI feedback becomes part of the social structure of learning, not a separate channel that students can choose to avoid.

The second refinement concerns language. The theoretical framework is largely built on literature from English-speaking and other monolingual contexts. It assumes that AI tools will operate in a single language. The focus group highlighted that Kyrgyz schools use several languages: Kyrgyz, Russian, Uzbek, Tajik and Dungan. There are also regional differences in vocabulary, even within Kyrgyz. Participants were clear that an AI system that speaks only one of these languages will leave some students behind. This is not only a technical challenge. It is also an equity issue. If AI Grader is to be used as a tool to close feedback gaps, it must not create new gaps based on language. This implies that language coverage and language accuracy need to be treated as design priorities. It also implies that evaluation of the system must include checks on performance across the different languages and dialects present in the school system.

The third refinement concerns how teacher trust develops over time. The theoretical account of the HITL model emphasizes that teachers should remain in charge of assessment decisions. It does not describe the psychological process through which teachers come to trust an AI partner. Participant 5's explanation fills this gap. She plans to check AI outputs against her own marking for one to three weeks. If the system proves reliable over that period, she expects that trust will develop. Trust here is not assumed. It is earned through repeated, verifiable success. This suggests that early deployment of AI Grader should include a phase in which teachers mark a sample of work manually and then compare their judgments with the AI's. The goal is not to prove that the AI is perfect, but to give teachers a concrete basis for calibration. This may reduce anxiety and make it more likely that teachers will feel comfortable using the system as part of their practice.

5.4 Implications for the AI Grader Design

The findings from the focus group have several direct implications for how the AI Grader scenarios should be implemented and refined.

First, Scenario 3 should be the starting point for deployment in Kyrgyz schools. It is the most widely supported scenario. It aligns directly with teachers

expressed need for better diagnostic information. Keeping tier labels invisible to students remains important. This avoids turning readiness levels into fixed ability labels.

Second, Scenario 2 should be closely tied to teacher practice in the classroom. AI-generated feedback should not travel alone by email. Instead, it should be followed by structured activities that require students to engage with the feedback. Whole-class discussions of typical errors and targeted re-do tasks are examples of such activities. A simple workflow is possible. The AI produces feedback and a class-wide error summary. The teacher reviews a small sample to check quality. The following day, the teacher selects one or two common error types and uses them as the focus for a short mini-lesson. This pattern keeps the human-in-the-loop principle intact and also addresses concerns about motivation and dependency.

Third, Scenario 1 needs a sharper value proposition. If teachers see it only as a slower way to collect answers, they will prefer existing tools such as Google Forms. The scenario must therefore emphasize its unique contribution. This is the analysis of intermediate steps in student working, not just final answers. The system can identify where a solution path breaks and why. This is not something a simple quiz platform can do. If this capability is demonstrated clearly to teachers, and if the workflow is optimized to reduce friction, Scenario 1 may gain more acceptance.

Finally, all three scenarios require two enabling conditions in the Kyrgyz context. The first is a multilingual deployment strategy. The second is a structured pilot-and-verify phase for teachers. A multilingual strategy ensures that students in different language streams can all benefit from improved feedback. A pilot-and-verify phase allows teachers to build trust in the system through their own professional judgment. Together, these conditions increase the likelihood that AI Grader will be both pedagogically effective and professionally acceptable in real classrooms.

6 Conclusion

This thesis investigated the pedagogical potential of Large Language Models in implementing personalized feedback systems in secondary school mathematics education. The study was motivated by a well-documented structural problem: teachers in secondary mathematics classrooms cannot consistently provide the quality and frequency of feedback that research identifies as most effective for student learning.

The work proceeded in three stages, structured around the research objectives stated in Section~1.3.

The first stage was a systematic review of relevant literature. The review examined formative assessment theory, personalized learning, differentiated instruction, Vygotsky’s Zone of Proximal Development, scaffolding pedagogy, self-regulated learning, and the human-in-the-loop model. Together, these frameworks form the Pedagogical Justification Framework for AI-Assisted Mathematics Assessment, which defines the design criteria for the AI Grader system.

The second stage was system design. Three concrete implementation scenarios were developed: Scenario 1 (end-of-lesson exit tickets for immediate formative assessment), Scenario 2 (daily homework review and feedback automation), and Scenario 3 (pre-lesson diagnostic support for differentiated instruction). Each scenario was designed to operationalize specific combinations of the theoretical principles identified in the literature review.

The third stage was empirical validation. A focus group with six practicing secondary school mathematics teachers in Kyrgyzstan was conducted to evaluate the scenarios, examine their alignment with real classroom conditions, and identify contextual factors not captured by the theoretical framework.

6.1 Answer to the Research Hypothesis

The thesis hypothesis states that LLM-based automated grading systems, when grounded in formative assessment theory and Vygotsky’s Zone of Proximal Development, can effectively reduce the feedback gap in secondary school mathematics. The hypothesis further states that such systems can provide timely, personalized, and process-oriented feedback at scale while maintaining teacher authority through a human-in-the-loop model.

The findings of the study support this hypothesis, with important qualifications. All six focus group participants confirmed that the feedback burden they experience is structural and cannot be resolved through individual effort. This confirms the diagnosis on which the hypothesis rests. The teachers also endorsed the AI Grader framework as a plausible response. They recognized the potential of automated diagnostics and feedback to address workload, improve precision, and support differentiated planning.

At the same time, the hypothesis is supported conditionally rather than unconditionally. The focus group identified three conditions that must be met for the hypothesis to hold in practice. First, the system must be embedded within teacher-led instructional structures, not delivered as a standalone automated service. Schaaf et al. (2026) show that AI tool use alone does not predict learning gains; what matters is how the tool is integrated into classroom routines. Second, the system must preserve and actively support teacher authority. Teachers in the focus group drew a clear boundary between what AI can do and what must remain with the teacher, particularly in the relational and motivational dimensions of learning. Third, the system must be appropriate for the specific institutional and linguistic context. In Kyrgyzstan, this means multilingual functionality and a trust-building onboarding process.

Subject to these conditions, the hypothesis is supported. LLM-based systems can reduce the feedback gap. They cannot do so independently of the pedagogical and institutional context in which they are deployed.

6.2 Contributions of the Study

The study makes three interconnected contributions.

The first is theoretical. The Pedagogical Justification Framework synthesizes six established educational theories into a coherent set of design criteria for AI-assisted mathematics assessment. This framework does not describe AI in education in general. It defines the specific conditions under which AI-generated feedback can be pedagogically effective: when it is formative rather than evaluative, calibrated to the learner's ZPD, designed to scaffold rather than replace independent thinking, and governed by a teacher who retains final authority over assessment decisions.

The second contribution is practical. The three AI Grader scenarios translate the theoretical framework into concrete classroom workflows. Each scenario

specifies the roles of the AI, the teacher, and the student. Each includes explicit human-in-the-loop checkpoints. Each is designed to address a specific pedagogical need identified in the literature review. The focus group provided validation and refinement of these designs. Scenario 3 was endorsed unanimously as the appropriate entry point for deployment. Scenarios 1 and 2 were validated with specific design recommendations for improving student engagement and teacher integration.

The third contribution is contextual. The study is, to the author's knowledge, among the first to examine AI-assisted mathematics feedback specifically in the Kyrgyzstan school context. The focus group identified three local requirements that the international literature does not address: multilingual AI functionality across the languages of instruction used in Kyrgyz schools; compatibility with institutional phone-ban policies; and a structured, iterative trust-building process for teacher adoption. These findings are relevant both for the development of the AI Grader system and for any future research on AI in education in Central Asian school contexts.

6.3 Limitations

Several limitations should be acknowledged.

The empirical phase relied on a single focus group with six participants. This is appropriate for exploratory qualitative research, but the findings cannot be generalized to secondary mathematics teachers in Kyrgyzstan as a whole. The results are indicative. They describe the perspectives of a specific group of practitioners in a specific institutional context.

The scenarios were evaluated as proposed designs, not as deployed systems. Participants offered prospective judgments about usefulness, trust, and feasibility. These judgments may not fully predict how teachers and students would respond to the system in actual use. Real deployment could surface challenges, benefits, and unexpected uses that a design-evaluation study cannot anticipate.

The focus group format may have introduced some degree of social desirability. Participants may have expressed more positive views about AI than they would hold privately, particularly in a group setting with a moderator presenting a specific system for evaluation. The moderator's explicit encouragement of disagreement was intended to reduce this risk, but cannot eliminate it entirely.

Finally, the AI Grader system described in this thesis is a conceptual framework. The technical components, including OCR accuracy, LLM prompt design, and feedback delivery infrastructure, remain to be fully implemented and tested. The pedagogical validity of the framework depends on technical reliability. A system that produces inaccurate or inconsistent feedback may cause harm by undermining student understanding and eroding teacher trust, regardless of how well it is theoretically grounded.

6.4 Directions for Future Research

Several directions for future research emerge from the findings.

The most pressing priority is empirical testing of the deployed system. A pilot study in which the AI Grader is used in real classrooms would allow researchers to assess whether the feedback it produces is accurate, whether teachers integrate it as expected, and whether it produces measurable improvements in formative assessment quality. A controlled design comparing classrooms with and without the AI Grader would provide the most informative evidence.

A second direction concerns the development of trust over time. The focus group showed that teacher trust in AI systems is not granted upfront. It develops gradually through verification and experience. Longitudinal research tracking how teacher confidence in AI-generated feedback evolves across a full school term would provide important design guidance for the onboarding and professional development components of the system.

A third direction concerns the student experience. This thesis focused on teacher perspectives. Students are the primary recipients of AI-generated feedback. Research on how different student groups respond to AI feedback, including lower-attaining students, students who are reluctant to engage, and students across different language backgrounds, would extend and deepen the framework's empirical basis.

Finally, the multilingual requirement identified by the focus group calls for dedicated technical and linguistic research. Evaluating the accuracy of OCR and LLM feedback generation across the languages and regional varieties spoken in Kyrgyz schools is a prerequisite for equitable deployment. This work requires collaboration between educators, linguists, and AI developers, and could contribute to a wider literature on AI in multilingual, non-Western educational contexts.

As the author of this thesis and as a practicing mathematics teacher, this research has also served as a form of professional reflection. It helped me reconsider feedback not merely as a routine part of checking student work, but as a central pedagogical action that shapes how students understand their mistakes, regulate their learning, and build confidence in mathematics. The study also made clear that technology cannot replace the professional judgment, emotional presence, and responsibility of the teacher. Instead, the value of Large Language Models lies in their potential to support teachers in tasks that are often limited by time and workload, while leaving key pedagogical decisions in human hands. For my own teaching practice, this means that AI tools should be introduced carefully, critically, and purposefully: not as a shortcut, but as an instrument for making feedback more timely, individualized, and meaningful for students.

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7 Appendix

7.1 Appendix A: Early-generation feedback report (baseline prototype)

This appendix reproduces a complete feedback report produced by an **early** build of the AI Grader prototype. It is included not as an example of final quality, but as **baseline documentation**: it shows what automated output looked like when the pipeline prioritized end-to-end scoring, concise narrative feedback, and a stable PDF layout over extended remediation content.

Отчет:

test assignment short

Результат: 67 / 100

□ Общая обратная связь

Привет! Ты хорошо справился с первыми двумя задачами на вычисления. В третьей задаче решение отсутствует, поэтому над ней нужно поработать. Уверен, что разобравшись с процентами, ты легко с ней справишься в следующий раз.

□ Что получилось хорошо

- Верно выполнил сложение дробей
- Правильно вычислил значение выражения с десятичными дробями, несмотря на возможную опечатку в записи

□ Над чем поработать

- Внимательно проверяй, все ли задачи решены до конца
- Потренируйся в решении задач на последовательное изменение величины на процент

□ Подробный разбор задач

Ассистент учителя математики: Я проанализировал решения задач, подготовленные учеником, и подготовил подробный разбор.

Шаг 1-4: Анализ задач

Задача 1: Вычислить $\frac{3}{4} + \frac{5}{6}$ (Макс. баллов: 1) **Ученик написал:** $\frac{3}{4} + \frac{5}{6} = 1\frac{7}{12}$ **Моё решение:** Находим общий знаменатель 12. $\frac{3}{4} = \frac{9}{12}$, $\frac{5}{6} = \frac{10}{12}$. Сумма: $\frac{9}{12} + \frac{10}{12} = \frac{19}{12} = 1\frac{7}{12}$. **Вердикт:** □ **ВЕРНО Начислено баллов: 1 Причина:** Решение и ответ ученика полностью верны.

Задача 2: Найти значение выражения $2,5 \cdot 0,4 - 1,2$ (Макс. баллов: 1) **Ученик написал:** $2,5 \cdot 0,4 - 1,2 = -0,2$ **Моё решение:** $2,5 \cdot 0,4 = 1,0$. Затем $1,0 - 1,2 = -0,2$. **Вердикт:** □ **ВЕРНО Начислено баллов: 1 Причина:** Несмотря на опечатку в записи ($2,5 \cdot 0,4$ вместо $2,5 \cdot 0,4$), конечный числовой ответ верный. OCR мог некорректно распознать знак умножения.

Задача 3: Цена товара была 1200 сом. Её повысили на 15%, а затем снизили на 15%. Какой стала цена? (Макс. баллов: 1) **Ученик написал:** $\frac{1200}{1}$ **Моё решение:** Повышение на 15%: $1200 \cdot 1.15 = 1380$ сом. Снижение результата на 15%: $1380 \cdot 0.85 = 1173$ сома. **Вердикт:** □ **НЕ РЕШАЛ Начислено баллов: 0 Причина:** Ученик не предоставил ответа, только начало записи. Задача не решена.

Шаг 5: Подсчёт итогового балла

ПОДСЧЁТ БАЛЛОВ: - Сумма набранных баллов: 2 - Максимально возможная сумма: 3 - Итоговый процент = $(2 / 3) \times 100 \approx 66.67\%$

Шаг 6: JSON

7.2 Appendix B: Later-generation feedback report (expanded process-level feedback)

This appendix reproduces a feedback report from a **later** iteration of the prototype. The intent is to illustrate how the same underlying architecture (ingestion → OCR → graded comparison against a teacher-authored reference → PDF notification) can produce substantially richer **formative** output when assignment metadata, prompts, and reporting templates encode stronger pedagogical intent.

Результаты:

Умножение десятичных дробей

70 / 100

Оценка: 4 / 5

Общая обратная связь

Вы хорошо справились с большинством заданий по умножению десятичных дробей. Вам удалось продемонстрировать понимание процесса умножения столбиком. Обратите внимание на точность в деталях, это поможет Вам достичь идеальных результатов.

Что получилось хорошо

- Верное выполнение умножения десятичных дробей на натуральные числа и другие десятичные дроби
- Аккуратное оформление решений столбиком

Над чем поработать

- Внимательнее переписывайте условия задач, чтобы избежать ошибок, как в задаче 3.
- Тщательнее проверяйте положение десятичной запятой в окончательном ответе, особенно при умножении на натуральные числа, как в задаче 5.

Рекомендуемые материалы

- [Нужно повторить умножение столбиком](#)
- [Нужно повторить умножение десятичных дробей](#)

Задания для дополнительной практики

1. Решите столбиком эти примеры $4,2 * 3$
2. Решите столбиком эти примеры $6,7 * 0,3$
3. Решите столбиком эти примеры $0,22 * 0,06$

Разбор задач

Задача 1

Умножить 5,8 на 3.

Правильное решение:

- Умножаем 58 на 3: $58 \times 3 = 174$.
- В числе 5,8 один знак после запятой.
- В результате должен быть один знак после запятой.
- $5,8 \times 3 = 17,4$.

Ответ ученика: $5,8 * 3 = 17,4$

Результат: Верно

Баллы: 1 / 1

Задача 2

Умножить 7,8 на 0,6.

Правильное решение:

- Умножаем 78 на 6: $78 \times 6 = 468$.
- В числе 7,8 один знак после запятой, в числе 0,6 один знак после запятой.
- Всего $1 + 1 = 2$ знака после запятой.
- В результате должно быть два знака после запятой.
- $7,8 \times 0,6 = 4,68$.

Ответ ученика: $7,8 * 0,6 = 4,68$

Результат: Верно

Баллы: 1 / 1

Задача 3

Умножить 0,25 на 0,04.

Правильное решение:

- Умножаем 25 на 4: $25 \times 4 = 100$.
- В числе 0,25 два знака после запятой, в числе 0,04 два знака после запятой.
- Всего $2 + 2 = 4$ знака после запятой.
- В результате должно быть четыре знака после запятой.
- $0,25 \times 0,04 = 0,0100 = 0,01$.

Ответ ученика: $0,25 * 0,4 = 0,100$

Результат: Неверно

Где ошибка: Вы допустили ошибку в условии задачи, умножив на 0,4 вместо 0,04. Это привело к неверному результату.

Баллы: 0 / 1

Задача 4

Умножить 0,125 на 0,5.

Правильное решение:

- Умножаем 125 на 5: $125 \times 5 = 625$.
- В числе 0,125 три знака после запятой, в числе 0,5 один знак после запятой.
- Всего $3 + 1 = 4$ знака после запятой.
- В результате должно быть четыре знака после запятой.
- $0,125 \times 0,5 = 0,0625$.

Ответ ученика: $0,125 \times 0,5 = 0,0625$

Результат: Верно

Баллы: 1 / 1

Задача 5

Умножить 37,5 на 4.

Правильное решение:

- Умножаем 375 на 4: $375 \times 4 = 1500$.
- В числе 37,5 один знак после запятой.
- В результате должен быть один знак после запятой.
- $37,5 \times 4 = 150,0 = 150$.

Ответ ученика: $37,5 \times 4 = 15,00$

Результат: Частично верно

Где ошибка: Вы верно выполнили умножение, но допустили ошибку в постановке десятичной запятой. 15,00 не равно 150.

Баллы: 0.5 / 1

Итоговый балл

Показатель	Значение
Задача 1	1 / 1
Задача 2	1 / 1
Задача 3	0 / 1
Задача 4	1 / 1
Задача 5	0.5 / 1
Итого	3.5 / 5
Процент	70%
Итоговая оценка	4 / 5

7.3 Appendix C: Links to online material

This appendix lists supplementary web-hosted materials that accompany the study: an anonymized data repository, a student-facing tutorial for using the AI Grader workflow, and an example of how Google Forms were integrated into assignment intake. These links are not substitutes for the methodological detail in Chapter 3; they document **how participants encountered the system in practice** and support transparency for readers who wish to inspect artifacts beyond what the printed thesis can embed.

The links are as follows:

- Shared folder (data and supplementary materials)
- AIsuluu: student tutorial (AI Grader protocol)
- AIsuluu: Google Form examples